50. Pulse Density Modulation Interface (PDM-IF)

50.1 Overview

The Pulse Density Modulation Interface (PDM-IF) has maximum three channels that are connectable with external microphone which outputs the pulse density modulated signal. PDM-IF is connectable with up to three external microphones. PDM-IF can filter and convert 1-bit digital data streams that were pulse density modulated at a high sampling rate into 20-bit or 16-bit digital data at a lower sampling rate.

In this section, PCLK refers to PCLKB.

Table 50.1 lists the specifications of PDM-IF. Figure 50.1 shows the block diagrams of PDM-IF.

Table 50.1 PDM-IF specification

Item	Description
Number of channels	3 channels

Functions	• Capable of filtering 1-bit digital input data PDM_DATAn (n = 0 to 2) and converting them into 20-bit or 16-bit digital data. The bit order is little-endian.
	• IP supports stereo microphone (L/R sampling by rising/falling clock edge).
	• IP supports sound activity detector.
	• Each channel of IP includes programmable filters: 4th order sinc filter, high-pass filter (for suppression of DC bias), compensation filter (for sinc passband distortion), half-band decimation filter (for aliasing distortion).
	  – IP supports programmable and flexible decimation ratios.
	  – The sinc filter is selectable as first-, second-, third-, fourth-order.
	• IP supports DMA operation through APB.
	• Each channel of IP has an internal buffer.
	  – Buffer size as 32 stages at design time
	  – Capable of storing voice data during low power mode
	• Error detection functions can be used for debugging.

Interrupt sources	Maximum 7 sources 1. Data reception interrupt per channel (Maximum 3 interrupts) 2. Sound detection interrupt (shared between channels) 3. Error detection interrupt per channel (Maximum 3 interrupts)
Low power consumption function	• IP supports microphone low power configuration (slower clock). IP can switch to higher clock speed after sound detection recognized. • MCU can stop bus clock during low power mode.
Bus interface	• IP has an APB interface which is confirmed to APB4.
TrustZone filter	• Security and Privilege attribution can be set.

Each channel of PDM-IF consists of following blocks:

- Clock control: It generates PDMCLKn to microphone, and internal timing signal to sinc filter.
- Data input control: It receives PDMDATAn from microphone.
- Sinc filter: It converts the 1-bit digital data stream input from PDMDATAn to 34-bit signed data. And then the 34-bit signed data is clipped to 20-bit signed data.
- High-pass filter: It is a filter for suppression of DC bias.
- Compensation filter: It is a filter for sinc passband distortion.
- Low-pass (half-band decimation) filter: It is a filter for aliasing distortion.
- Moving average filter: It is a simple filter for noise reduction before sound detection.
- Sound detection: It generates the wakeup interrupt to recover from low power mode.
- Data buffer: It stores 20-bit or 16-bit data filtered by some filters.
- Short circuit detection: It detects the short circuit error which indicates consecutive 0's or 1's input from PDMDATAn.
- Overvoltage detection: It detects the overvoltage error.
- Error control: It generates error flags from other blocks.

The 1-bit digital data stream input from PDMDATAn pin (n = 0 to 2) is converted to 34-bit signed data by the sinc filter and then the data is clipped to 20-bit signed data. The 20-bit signed data is used for sound detection and data storing in buffer.

Sound detection block receives 20-bit signed data through high-pass filter block and moving average filter block. The block compares received data to the programmed thresholds and generates a wakeup interrupt in case that the data is above the upper threshold or below the lower threshold.

Data buffer block receives 20-bit signed data through high-pass filter, compensation filter, and low-pass filter. The block stores the 20-bit signed data or the clipped 16-bit signed data. And then the block sends the 20-bit signed data or 16-bit signed data to APB-IF once APB read access occurs.

Table 50.2 PDM I/O pins (n = 0 to 2)

Function	Pin name	I/O	Description
PDM-IF	PDMCLKn	Output	Output pins for clock
	PDMDATAn	Input	Input pins for data

Figure 50.1 shows the main components of PDM-IF.

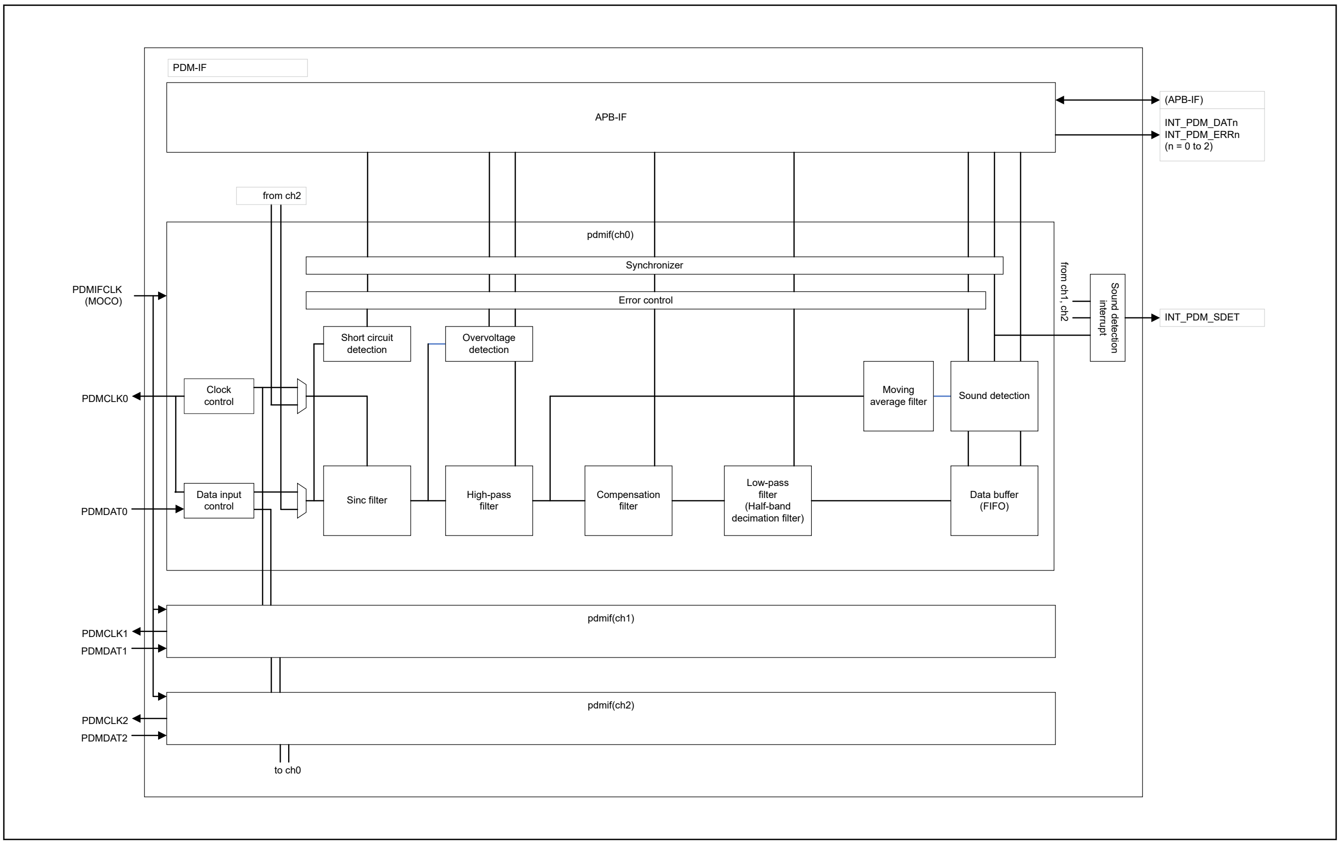


Figure 50.1 PDM-IF block diagram

50.2 Register Description

50.2.1 Register Map

Table 50.3 Register map (1 of 5)

Base address symbol	Offset address	Symbol	Register name	R/W	Initial value	Writable period*1				
						(1) Start flow	(2) Stop flow	(3) Normal processing flow	(4) Low power mode transition flow	(5) Normal mode transition flow
PDMIF	0x000	PDCSTRTR	Channel Software Start Trigger Register	W	0x00000000	✔	—	—	—	—
PDMIF	0x004	PDCSTPTR	Channel Software Stop Trigger Register	W	0x00000000	—	✔	—	—	—
PDMIF	0x008	PDCCHGTR	Channel Software Change Trigger Register	W	0x00000000	—	—	—	✔	✔
PDMIF	0x00C	PDCICR	Channel Interrupt Control Register	R/W	0x00000000	✔	✔	—	✔	✔
PDMIF	0x010	PDCSR	Channel Status Register	R	0x00000000	—	—	—	—	—
PDMIF	0x014	PDCSCR	Channel Status Clear Register	W	0x00000000	✔	✔	—	✔	✔
PDMIF	0x020	PDCSDCR	Channel Sound Detection Control Register	R/W	0x00000000	✔	✔	—	✔	✔
PDMIF	0x024	PDCDRCR	Channel Data Read Control Register	R/W	0x00000000	—	—	✔	—	—
PDMIF	0x028	PDCDCR	Channel Data Clear Register	W	0x00000000	—	—	✔*2	—	—
PDMIF	0x080	PDVR	Version Register	R	0x00000010	—	—	—	—	—
PDMIF	0x100 + 0x100 x n	PDSTRTRCHn	Software Start Trigger Register Channel n (n = 0 to 2)	W	0x00000000	✔	—	—	—	—
PDMIF	0x104 + 0x100 x n	PDSTPTRCHn	Software Stop Trigger Register Channel n (n = 0 to 2)	W	0x00000000	—	✔	—	—	—
PDMIF	0x108 + 0x100 x n	PDCHGTRCHn	Software Change Trigger Register Channel n (n = 0 to 2)	W	0x00000000	—	—	—	✔	✔
PDMIF	0x10C + 0x100 x n	PDICRCHn	Interrupt Control Register Channel n (n = 0 to 2)	R/W	0x00000000	✔	✔	—	✔	✔
PDMIF	0x110 + 0x100 x n	PDSDCRCHn	Status Detection Control Register Channel n (n = 0 to 2)	R/W	0x00000000	✔	✔	—	—	—
PDMIF	0x114 + 0x100 x n	PDSRCHn	Status Register Channel n (n = 0 to 2)	R	0x00000000	—	—	—	—	—

Table 50.3 Register map (2 of 5)

Base address symbol	Offset address	Symbol	Register name	R/W	Initial value	Writable period[*1]				
						(1) Start flow	(2) Stop flow	(3) Normal processing flow	(4) Low power mode transition flow	(5) Normal mode transition flow
PDMIF	0x118 + 0x100 x n	PDSCRCHn	Status Clear Register Channel n (n = 0 to 2)	W	0x00000000	✔	✔	—	✔	✔
PDMIF	0x120 + 0x100 x n	PDMDSRCHn	Mode Setting Register Channel n (n = 0 to 2)	R/W	0x00000000	✔	—	—	—	—
PDMIF	0x124 + 0x100 x n	PDSFCRCHn	Sinc filter Control Register Channel n (n = 0 to 2)	R/W	0x057C0000	✔	—	—	✔	✔
PDMIF	0x128 + 0x100 x n	PDHFCS0RCHn	High-pass filter Coefficient s(0) Register Channel n (n = 0 to 2)	R/W	0x00003F61	✔	—	—	—	—
PDMIF	0x12C + 0x100 x n	PDHFCK1RCHn	High-pass filter Coefficient k(1) Register Channel n (n = 0 to 2)	R/W	0x00003EC1	✔	—	—	—	—
PDMIF	0x130 + 0x100 x n	PDHFCH0RCHn	High-pass filter Coefficient h(0) Register Channel n (n = 0 to 2)	R/W	0x00004000	✔	—	—	—	—
PDMIF	0x134 + 0x100 x n	PDHFCH1RCHn	High-pass filter Coefficient h(1) Register Channel n (n = 0 to 2)	R/W	0x0000C000	✔	—	—	—	—
PDMIF	0x138 + 0x100 x n	PDCFCH00RCHn	Compensation filter Coefficient h(0) Register Channel n (n = 0 to 2)	R/W	0x00001FE8	✔	—	—	—	—
PDMIF	0x13C + 0x100 x n	PDCFCH01RCHn	Compensation filter Coefficient h(1) Register Channel n (n = 0 to 2)	R/W	0x00000039	✔	—	—	—	—
PDMIF	0x140 + 0x100 x n	PDCFCH02RCHn	Compensation filter Coefficient h(2) Register Channel n (n = 0 to 2)	R/W	0x0000003C	✔	—	—	—	—
PDMIF	0x144 + 0x100 x n	PDCFCH03RCHn	Compensation filter Coefficient h(3) Register Channel n (n = 0 to 2)	R/W	0x00001E56	✔	—	—	—	—
PDMIF	0x148 + 0x100 x n	PDCFCH04RCHn	Compensation filter Coefficient h(4) Register Channel n (n = 0 to 2)	R/W	0x000001DC	✔	—	—	—	—
PDMIF	0x14C + 0x100 x n	PDCFCH05RCHn	Compensation filter Coefficient h(5) Register Channel n (n = 0 to 2)	R/W	0x000006E1	✔	—	—	—	—
PDMIF	0x150 + 0x100 x n	PDCFCH06RCHn	Compensation filter Coefficient h(6) Register Channel n (n = 0 to 2)	R/W	0x000001DC	✔	—	—	—	—

Table 50.3 Register map (3 of 5)

Base address symbol	Offset address	Symbol	Register name	R/W	Initial value	Writable period*1				
						(1) Start flow	(2) Stop flow	(3) Normal processing flow	(4) Low power mode transition flow	(5) Normal mode transition flow
PDMIF	0x154 + 0x100 x n	PDCFCH07RCHn	Compensation filter Coefficient h(7) Register Channel n (n = 0 to 2)	R/W	0x00001E56	✔	—	—	—	—
PDMIF	0x158 + 0x100 x n	PDCFCH08RCHn	Compensation filter Coefficient h(8) Register Channel n (n = 0 to 2)	R/W	0x0000003C	✔	—	—	—	—
PDMIF	0x15C + 0x100 x n	PDCFCH09RCHn	Compensation filter Coefficient h(9) Register Channel n (n = 0 to 2)	R/W	0x00000039	✔	—	—	—	—
PDMIF	0x160 + 0x100 x n	PDCFCH10RCHn	Compensation filter Coefficient h(10) Register Channel n (n = 0 to 2)	R/W	0x00001FE8	✔	—	—	—	—
PDMIF	0x164 + 0x100 x n	PDLFCH010RCHn	Low-pass filter Coefficient h0(10) Register Channel n (n = 0 to 2)	R/W	0x00000400	✔	—	—	—	—
PDMIF	0x168 + 0x100 x n	PDLFCH100RCHn	Low-pass filter Coefficient h1(0) Register Channel n (n = 0 to 2)	R/W	0x00001FF8	✔	—	—	—	—
PDMIF	0x16C + 0x100 x n	PDLFCH101RCHn	Low-pass filter Coefficient h1(1) Register Channel n (n = 0 to 2)	R/W	0x0000000A	✔	—	—	—	—
PDMIF	0x170 + 0x100 x n	PDLFCH102RCHn	Low-pass filter Coefficient h1(2) Register Channel n (n = 0 to 2)	R/W	0x00001FF0	✔	—	—	—	—
PDMIF	0x174 + 0x100 x n	PDLFCH103RCHn	Low-pass filter Coefficient h1(3) Register Channel n (n = 0 to 2)	R/W	0x00000018	✔	—	—	—	—
PDMIF	0x178 + 0x100 x n	PDLFCH104RCHn	Low-pass filter Coefficient h1(4) Register Channel n (n = 0 to 2)	R/W	0x00001FDC	✔	—	—	—	—
PDMIF	0x17C + 0x100 x n	PDLFCH105RCHn	Low-pass filter Coefficient h1(5) Register Channel n (n = 0 to 2)	R/W	0x00000034	✔	—	—	—	—
PDMIF	0x180 + 0x100 x n	PDLFCH106RCHn	Low-pass filter Coefficient h1(6) Register Channel n (n = 0 to 2)	R/W	0x00001FB3	✔	—	—	—	—
PDMIF	0x184 + 0x100 x n	PDLFCH107RCHn	Low-pass filter Coefficient h1(7) Register Channel n (n = 0 to 2)	R/W	0x00000076	✔	—	—	—	—

Table 50.3 Register map (4 of 5)

Base address symbol	Offset address	Symbol	Register name	R/W	Initial value	Writable period[*1]				
						(1) Start flow	(2) Stop flow	(3) Normal processing flow	(4) Low power mode transition flow	(5) Normal mode transition flow
PDMIF	0x188 + 0x100 x n	PDLFCH108RCHn	Low-pass filter Coefficient h1(8) Register Channel n (n = 0 to 2)	R/W	0x00001F2E	✔	—	—	—	—
PDMIF	0x18C + 0x100 x n	PDLFCH109RCHn	Low-pass filter Coefficient h1(9) Register Channel n (n = 0 to 2)	R/W	0x00000289	✔	—	—	—	—
PDMIF	0x190 + 0x100 x n	PDLFCH110RCHn	Low-pass filter Coefficient h1(10) Register Channel n (n = 0 to 2)	R/W	0x00000289	✔	—	—	—	—
PDMIF	0x194 + 0x100 x n	PDLFCH111RCHn	Low-pass filter Coefficient h1(11) Register Channel n (n = 0 to 2)	R/W	0x00001F2E	✔	—	—	—	—
PDMIF	0x198 + 0x100 x n	PDLFCH112RCHn	Low-pass filter Coefficient h1(12) Register Channel n (n = 0 to 2)	R/W	0x00000076	✔	—	—	—	—
PDMIF	0x19C + 0x100 x n	PDLFCH113RCHn	Low-pass filter Coefficient h1(13) Register Channel n (n = 0 to 2)	R/W	0x00001FB3	✔	—	—	—	—
PDMIF	0x1A0 + 0x100 x n	PDLFCH114RCHn	Low-pass filter Coefficient h1(14) Register Channel n (n = 0 to 2)	R/W	0x00000034	✔	—	—	—	—
PDMIF	0x1A4 + 0x100 x n	PDLFCH115RCHn	Low-pass filter Coefficient h1(15) Register Channel n (n = 0 to 2)	R/W	0x00001FDC	✔	—	—	—	—
PDMIF	0x1A8 + 0x100 x n	PDLFCH116RCHn	Low-pass filter Coefficient h1(16) Register Channel n (n = 0 to 2)	R/W	0x00000018	✔	—	—	—	—
PDMIF	0x1AC + 0x100 x n	PDLFCH117RCHn	Low-pass filter Coefficient h1(17) Register Channel n (n = 0 to 2)	R/W	0x00001FF0	✔	—	—	—	—
PDMIF	0x1B0 + 0x100 x n	PDLFCH118RCHn	Low-pass filter Coefficient h1(18) Register Channel n (n = 0 to 2)	R/W	0x0000000A	✔	—	—	—	—
PDMIF	0x1B4 + 0x100 x n	PDLFCH119RCHn	Low-pass filter Coefficient h1(19) Register Channel n (n = 0 to 2)	R/W	0x00001FF8	✔	—	—	—	—
PDMIF	0x1B8 + 0x100 x n	PDSDLTRCHn	Sound Detection Lower Threshold Register Channel n (n = 0 to 2)	R/W	0x00000000	✔	—	—	✔	—

Table 50.3 Register map (5 of 5)

Base address symbol	Offset address	Symbol	Register name	R/W	Initial value	Writable period*1				
						(1) Start flow	(2) Stop flow	(3) Normal processing flow	(4) Low power mode transition flow	(5) Normal mode transition flow
PDMIF	0x1BC + 0x100 x n	PDSDUTRCHn	Sound Detection Upper Threshold Register Channel n (n = 0 to 2)	R/W	0x00000000	✔	—	—	✔	—
PDMIF	0x1C0 + 0x100 x n	PDDBCRCHn	Data Buffer Control Register Channel n (n = 0 to 2)	R/W	0x00000000	✔	—	—	—	—
PDMIF	0x1C4 + 0x100 x n	PDSCTSRCHn	Short Circuit Threshold Setting Register Channel n (n = 0 to 2)	R/W	0x00000000	✔	—	—	—	—
PDMIF	0x1C8 + 0x100 x n	PDOVLTRCHn	Overvoltage Lower Threshold Register Channel n (n = 0 to 2)	R/W	0x00000000	✔	—	—	—	—
PDMIF	0x1CC + 0x100 x n	PDOVUTRCHn	Overvoltage Upper Threshold Register Channel n (n = 0 to 2)	R/W	0x00000000	✔	—	—	—	—
PDMIF	0x1E0 + 0x100 x n	PDDRCRCHn	Data Read Control Register Channel n (n = 0 to 2)	R/W	0x00000000	—	—	✔	—	—
PDMIF	0x1E4 + 0x100 x n	PDDCRCHn	Data Clear Register Channel n (n = 0 to 2)	W	0x00000000	—	—	✔*2	—	—
PDMIF	0x1E8 + 0x100 x n	PDDRRCHn	Data Read Register Channel n (n = 0 to 2)	R	0x00000000	—	—	—	—	—
PDMIF	0x1EC + 0x100 x n	PDDSRCHn	Data Status Register Channel n (n = 0 to 2)	R	0x00000000	—	—	—	—	—

Note: Some registers are mirroring together between base address symbol PDM_CTL and PDMn registers. See section 50.2.67. Mirroring Register.

Note 1. See section 50.4. Setting Flow.

Note 2. The register is not shown in the Setting flow, but it is allowed to use in the flow.

50.2.2 PDCSTRTR : Channel Software Start Trigger Register

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	STRTRG2	STRTRG1	STRTRG0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STRTRG0	Channel 0 Start Trigger 0: Do nothing 1: Start channel 0	W
1	STRTRG1	Channel 1 Start Trigger 0: Do nothing 1: Start channel 1	W
2	STRTRG2	Channel 2 Start Trigger 0: Do nothing 1: Start channel 2	W
31:3	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

50.2.3 PDCSTPTR : Channel Software Stop Trigger Register

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x04

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	STPTRG2	STPTRG1	STPTRG0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STPTRG0	Channel 0 Stop Trigger 0: Do nothing 1: Stop channel 0	W
1	STPTRG1	Channel 1 Stop Trigger 0: Do nothing 1: Stop channel 1	W
2	STPTRG2	Channel 2 Stop Trigger 0: Do nothing 1: Stop channel 2	W
31:3	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

50.2.4 PDCCHGTR : Channel Software Change Trigger Register

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x08

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	CHGTRG2	CHGTRG1	CHGTRG0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CHGTRG0	Channel 0 Change Trigger 0: Do nothing 1: Change clock (PDM_CLK0) setting	W
1	CHGTRG1	Channel 1 Change Trigger 0: Do nothing 1: Change clock (PDM_CLK1) setting	W
2	CHGTRG2	Channel 2 Change Trigger 0: Do nothing 1: Change clock (PDM_CLK2) setting	W
31:3	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

50.2.5 PDCICR : Channel Interrupt Control Register

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x0C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	IEDE2	IEDE1	IEDE0	—	—	—	—	—	IDRE2	IDRE1	IDRE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	ISDE2	ISDE1	ISDE0	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	ISDE0	Channel 0 Sound Detection Interrupt Enable Bit 0: Do not allow PDM_SDET interrupt to be issued when the sound for channel 0 is detected 1: Allow PDM_SDET interrupt to be issued when the sound for channel 0 is detected	R/W
9	ISDE1	Channel 1 Sound Detection Interrupt Enable Bit 0: Do not allow PDM_SDET interrupt to be issued when the sound for channel 1 is detected 1: Allow PDM_SDET interrupt to be issued when the sound for channel 1 is detected	R/W
10	ISDE2	Channel 2 Sound Detection Interrupt Enable Bit 0: Do not allow PDM_SDET interrupt to be issued when the sound for channel 2 is detected 1: Allow PDM_SDET interrupt to be issued when the sound for channel 2 is detected	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
16	IDRE0	Channel 0 Data Reception Interrupt Enable Bit 0: Do not allow PDM_DAT0 interrupt to be issued 1: Allow PDM_DAT0 interrupt to be issued	R/W
17	IDRE1	Channel 1 Data Reception Interrupt Enable Bit 0: Do not allow PDM_DAT1 interrupt to be issued 1: Allow PDM_DAT1 interrupt to be issued	R/W
18	IDRE2	Channel 2 Data Reception Interrupt Enable Bit 0: Do not allow PDM_DAT2 interrupt to be issued 1: Allow PDM_DAT2 interrupt to be issued	R/W
23:19	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
24	IEDE0	Channel 0 Error Detection Interrupt Enable Bit 0: Do not allow PDM_ERR0 interrupt to be issued 1: Allow PDM_ERR0 interrupt to be issued	R/W
25	IEDE1	Channel 1 Error Detection Interrupt Enable Bit 0: Do not allow PDM_ERR1 interrupt to be issued 1: Allow PDM_ERR1 interrupt to be issued	R/W
26	IEDE2	Channel 2 Error Detection Interrupt Enable Bit 0: Do not allow PDM_ERR2 interrupt to be issued 1: Allow PDM_ERR2 interrupt to be issued	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.6 PDCSR : Channel Status Register

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x10

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	EDF2	EDF1	EDF0	—	—	—	—	—	DRF2	DRF1	DRF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	SDF2	SDF1	SDF0	—	—	—	—	—	STATE2	STATE1	STATE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STATE0	Channel 0 State 0: Channel 0 stop 1: Channel 0 in operation	R
1	STATE1	Channel 1 State 0: Channel 1 stop 1: Channel 1 in operation	R
2	STATE2	Channel 2 State 0: Channel 2 stop 1: Channel 2 in operation	R
7:3	—	These bits are read as 0.	R
8	SDF0	Channel 0 Sound Detection Flag 0: Indicates that the sound which exceeded the threshold is not detected 1: Indicates that the sound which exceeded the threshold is detected	R
9	SDF1	Channel 1 Sound Detection Flag 0: Indicates that the sound which exceeded the threshold is not detected 1: Indicates that the sound which exceeded the threshold is detected	R
10	SDF2	Channel 2 Sound Detection Flag 0: Indicates that the sound which exceeded the threshold is not detected 1: Indicates that the sound which exceeded the threshold is detected	R
15:11	—	These bits are read as 0.	R
16	DRF0	Channel 0 Data Reception Flag 0: Indicates that the number of data stored in buffer does not exceed the threshold 1: Indicates that the number of data stored in buffer exceeded the threshold	R
17	DRF1	Channel 1 Data Reception Flag 0: Indicates that the number of data stored in buffer does not exceed the threshold 1: Indicates that the number of data stored in buffer exceeded the threshold	R

Bit	Symbol	Function	R/W
18	DRF2	Channel 2 Data Reception Flag 0: Indicates that the number of data stored in buffer does not exceed the threshold 1: Indicates that the number of data stored in buffer exceeded the threshold	R
23:19	—	These bits are read as 0.	R
24	EDF0	Channel 0 Error Detection Flag 0: Indicates that errors are not detected. 1: Indicates that errors are detected. See section 50.2.17. PDSRCHn : Status Register Channel n (n = 0 to 2) for details.	R
25	EDF1	Channel 1 Error Detection Flag 0: Indicates that errors are not detected. 1: Indicates that errors are detected. See section 50.2.17. PDSRCHn : Status Register Channel n (n = 0 to 2) for details.	R
26	EDF2	Channel 2 Error Detection Flag 0: Indicates that errors are not detected. 1: Indicates that errors are detected. See section 50.2.17. PDSRCHn : Status Register Channel n (n = 0 to 2) for details.	R
31:27	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

STATEn: See section 50.3.2. Channel Start and Channel Stop for details.

[Condition to become 0]

- When 1 is written to PDCSTPTR.STPTRGn.

[Condition to become 1]

- When 1 is written to PDCSTRTR.STRTRGn.

SDFn: See section 50.3.6. Sound Detection for details.

[Condition to become 0]

- When writing 1 to a flag clear register bit (PDCSCR.SDFCn).

[Condition to become 1]

- When PDCSDCR.SDEn = 1 and a moving average filter result is higher than or equal to PDSDUTR.SDETU[19:0].
- When PDCSDCR.SDEn = 1 and a moving average filter result is lower than or equal to PDSDLTR.SDETL[19:0].

DRFn: See section 50.3.7. Data Buffer for details.

[Condition to become 0]

- When PDCDRCR.DATREn = 0 or PDDSR.DATNUM[7:0] is lower than a threshold.

[Condition to become 1]

- When PDCDRCR.DATREn = 1 and PDDSR.DATNUM[7:0] is higher than or equal to a threshold which is configured by PDDBCR.DATRITHR[2:0].

EDFn

[Condition to become 0]

- When all error flags are cleared by writing to section 50.2.18. PDSCRCHn : Status Clear Register Channel n (n = 0 to 2), this flag is also cleared.

[Condition to become 1]

- When any error occurs. See section 50.2.17. PDSRCHn : Status Register Channel n (n = 0 to 2) register for the kind of error.

50.2.7 PDCSCR : Channel Status Clear Register

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x14

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	SDFC2	SDFC1	SDFC0	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	The write value should be 0.	W
8	SDFC0	Channel 0 Sound Detection Flag Clear 0: Do nothing 1: Clear PDCSR.SDF0	W
9	SDFC1	Channel 1 Sound Detection Flag Clear 0: Do nothing 1: Clear PDCSR.SDF1	W
10	SDFC2	Channel 2 Sound Detection Flag Clear 0: Do nothing 1: Clear PDCSR.SDF2	W
31:11	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

50.2.8 PDCSDCR : Channel Sound Detection Control Register

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x20

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SDE2	SDE1	SDE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SDE0	Channel 0 Sound Detection Enable Bit 0: Do not allow sound detection 1: Allow sound detection	R/W
1	SDE1	Channel 1 Sound Detection Enable Bit 0: Do not allow sound detection 1: Allow sound detection	R/W
2	SDE2	Channel 2 Sound Detection Enable Bit 0: Do not allow sound detection 1: Allow sound detection	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.9 PDCDRCR : Channel Data Read Control Register

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x24

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	DATRE2	DATRE1	DATRE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DATRE0	Channel 0 Data Read Enable Bit 0: Do not allow reading data from data buffer 1: Allow reading data from data buffer	R/W
1	DATRE1	Channel 1 Data Read Enable Bit 0: Do not allow reading data from data buffer 1: Allow reading data from data buffer	R/W
2	DATRE2	Channel 2 Data Read Enable Bit 0: Do not allow reading data from data buffer 1: Allow reading data from data buffer	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.10 PDCDCR : Channel Data Clear Register

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x28

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	DATC2	DATC1	DATC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DATC0	Channel 0 Data Clear 0: Do nothing 1: Clear data	W
1	DATC1	Channel 1 Data Clear 0: Do nothing 1: Clear data	W
2	DATC2	Channel 2 Data Clear 0: Do nothing 1: Clear data	W

Bit	Symbol	Function	R/W
31:3	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

50.2.11 PDVR : Version Register

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x80

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	VER[11:0]											
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	VER[11:0]	PDM-IF version is shown.	R
31:12	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

50.2.12 PDSTRTRCHn : Software Start Trigger Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x100 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STRTRG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STRTRG	Start Trigger 0: Do nothing 1: Start the channel	W
31:1	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

50.2.13 PDSTPTRCHn : Software Stop Trigger Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x104 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STPTRG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STPTRG	Stop Trigger 0: Do nothing 1: Stop the channel	W
31:1	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

50.2.14 PDCHGTRCHn : Software Change Trigger Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x108 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CHGTRG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CHGTRG	Change Trigger 0: Do nothing 1: Change settings of PDM_CLKn and sinc filter	W
31:1	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

50.2.15 PDICRCHn : Interrupt Control Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x10C + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	IEDE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	IDRE	ISDE	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	ISDE	Sound Detection Interrupt Enable Bit 0: Do not allow PDM_SDET interrupt to be issued 1: Allow PDM_SDET interrupt to be issued	R/W
2	IDRE	Data Reception Interrupt Enable Bit 0: Do not allow PDM_DATn interrupt to be issued 1: Allow PDM_DATn interrupt to be issued	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
16	IEDE	Error Detection Interrupt Enable Bit 0: Do not allow PDM_ERRn interrupt to be issued 1: Allow PDM_ERRn interrupt to be issued	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.16 PDSDCRCHn : Status Detection Control Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x110 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	BFOWDE	—	—	—	—	—	—	—	—	OVUDE	OVLDE	SCDE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SDE	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	SDE	Sound Detection Enable Bit 0: Do not allow sound detection 1: Allow sound detection	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W
16	SCDE	Short Circuit Detection Enable Bit 0: Do not allow detection of the PDM_DATAn pin short circuit 1: Allow detection of the PDM_DATAn pin short circuit	R/W

Bit	Symbol	Function	R/W
17	OVLDE	Overvoltage Lower Limit Exceeded Detection Enable Bit 0: Do not allow detection of data fallen lower limit 1: Allow detection of data fallen lower limit	R/W
18	OVUDE	Overvoltage Upper Limit Exceeded Detection Enable Bit 0: Do not allow detection of data exceeded upper limit 1: Allow detection of data exceeded upper limit	R/W
26:19	—	These bits are read as 0. The write value should be 0.	R/W
27	BFOWDE	Buffer Overwriting Detection Enable Bit 0: Do not allow detection of buffer overwriting 1: Allow detection of buffer overwriting	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.17 PDSRCHn : Status Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x114 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	BFOWDF	—	—	—	—	—	—	—	—	OVUDF	OVLDF	SCDF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	DRF	SDF	STATE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STATE	State 0: Channel stop 1: Channel in operation	R
1	SDF	Sound Detection Flag 0: Indicates that the sound which exceeded the threshold is not detected 1: Indicates that the sound which exceeded the threshold is detected	R
2	DRF	Data Reception Flag 0: Indicates that the number of data stored in buffer is lower than a threshold 1: Indicates that the number of data stored in buffer is higher than or equal to a threshold	R
15:3	—	These bits are read as 0.	R
16	SCDF	Short circuit detection flag. 0: Indicates that the PDM_DATAn pin short circuit is not detected. 1: Indicates that a short circuit on the PDM_DATAn pin has been detected.	R
17	OVLDF	Overvoltage Lower Limit Exceeded Detection Flag 0: Indicates that the data is not lower than the lower limit 1: Indicates that the data has fallen the lower limit	R
18	OVUDF	Overvoltage Upper Limit Exceeded Detection Flag 0: Indicates that the data has not exceeded the upper limit 1: Indicates that the data has exceeded the upper limit	R
26:19	—	These bits are read as 0.	R
27	BFOWDF	Buffer Overwriting Detection Flag 0: Indicates that buffer overwriting does not occur 1: Indicates that buffer overwriting occurs	R
31:28	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

STATE: For details, see section 50.3.2. Channel Start and Channel Stop.

[Condition to become 0]

- When 1 is written to PDSTPTR.STPTRG.

[Condition to become 1]

- When 1 is written to PDSTRTR.STRTRG.

SDF: For details, see section 50.3.6. Sound Detection.

[Condition to become 0]

- When writing 1 to the flag clear register bit (PDSCR.SDFC).

[Condition to become 1]

- When PDSDCR.SDE = 1 and a moving average filter result is PDSDUTR.SDETU[19:0] or higher.
- When PDSDCR.SDE = 1 and a moving average filter result is PDSDLTR.SDETL[19:0] or lower.

DRF: For details, see section 50.3.7. Data Buffer.

[Condition to become 0]

- When PDDRCR.DATRE = 0 or PDDSR.DATNUM[7:0] is lower than a threshold.

[Condition to become 1]

- When PDDRCR.DATRE = 1 and PDDSR.DATNUM[7:0] is higher than or equal to a threshold which is configured by PDDBCR.DATRITHR[2:0].

SCDF: For details, see section 50.3.8. Short-circuit Detection.

[Condition to become 0]

- When writing 1 to the flag clear register bit (PDSCR.SCDFC).

[Condition to become 1]

- When PDSDCR.SCDE = 1 and 1 is continuously input to the PDM_DATAn pin, exceeding the number of times set in PDSCTSR.SCDH[12:0].
- When PDSDCR.SCDE = 1 and 0 is continuously input to the PDM_DATAn pin, exceeding the number of times set in PDSCTSR.SCDL[12:0].

OVLDF: For details, see section 50.3.9. Overvoltage Detection.

[Condition to become 0]

- When writing 1 to the flag clear register bit (PDSCR.OVLDFC).

[Condition to become 1]

- When PDSDCR.OVLDE = 1 and a clipped sinc filter result is smaller than a lower limit (PDOVLTR.OVDL[19:0]).

OVUDF: For details, see section 50.3.9. Overvoltage Detection.

[Condition to become 0]

- When writing 1 to the flag clear register bit (PDSCR.OVUDFC).

[Condition to become 1]

- When PDSDCR.OVUDE = 1 and a clipped sinc filter result is larger than an upper limit (PDOVUTR.OVDU[19:0]).

BFOWDF: For details, see section 50.3.7. Data Buffer.

[Condition to become 0]

- When writing 1 to the flag clear register bit (PDSCR.BFOWDFC).

[Condition to become 1]

- When PDSDCR.BFOWDE = 1 and data buffer is full and data is written to buffer without reading.

50.2.18 PDSCRCHn : Status Clear Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x118 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	BFOWDFC	—	—	—	—	—	—	—	—	OVUDFC	OVLDFC	SCDFC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SDFC	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	The write value should be 0.	W
1	SDFC	Sound Detection Flag Clear 0: Do nothing 1: Clear PDSRCHn.SDF	W
15:2	—	The write value should be 0.	W
16	SCDFC	Short Circuit Detection Flag Clear 0: Do nothing 1: Clear PDSRCHn.SCDF	W
17	OVLDFC	Overvoltage Lower Limit Exceeded Detection Flag Clear 0: Do nothing 1: Clear PDSRCHn.OVLDF	W
18	OVUDFC	Overvoltage Upper Limit Exceeded Detection Flag Clear 0: Do nothing 1: Clear PDSRCHn.OVUDF	W
26:19	—	The write value should be 0.	W
27	BFOWDFC	Buffer Overwriting Detection Flag Clear 0: Do nothing 1: Clear PDSRCHn.BFOWDF	W
31:28	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

50.2.19 PDMDSRCHn : Mode Setting Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x120 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DBIS[3:0]				—	—	SDMAMD[1:0]		—	—	—	—	—	—	LFIS[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	CFIS[1:0]		—	—	HFIS[1:0]		—	SFMD[2:0]			—	—	—	INPSEL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	INPSEL	Input Data Select 0: Rise-edge data of channel n. 1: Fall-edge data of channel n-1. When n = 0, fall-edge data of channel 2 is selected.	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
6:4	SFMD[2:0]	Sinc Filter Mode Setting 0 0 1: 1-order 0 1 0: 2-order 0 1 1: 3-order Others: 4-order (default)	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
9:8	HFIS[1:0]	High-pass Filter Input Shift Setting 0 0: No shift 0 1: 1-bit right shift 1 0: 2-bit right shift 1 1: 3-bit right shift	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
13:12	CFIS[1:0]	Compensation Filter Input Shift Setting 0 0: No shift 0 1: 1-bit right shift 1 0: 2-bit right shift 1 1: 3-bit right shift	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
17:16	LFIS[1:0]	Low-pass (half-band decimation) Filter Input Shift Setting 0 0: No shift 0 1: 1-bit right shift 1 0: 2-bit right shift 1 1: 3-bit right shift	R/W
23:18	—	These bits are read as 0. The write value should be 0.	R/W
25:24	SDMAMD[1:0]	Moving Average Mode for Sound Detection Data 0 0: 1-order (filter is skipped) (default) 0 1: 2-order 1 0: 4-order 1 1: User prohibition	R/W
27:26	—	These bits are read as 0. The write value should be 0.	R/W
31:28	DBIS[3:0]	Data Buffer Input Shift Setting S: signed bit 0x0: 20-bit mode, {1{S}, [18:0]} 0x1: 20-bit mode, {2{S}, [18:1]} 0x2: 20-bit mode, {3{S}, [18:2]} 0x3: 20-bit mode, {4{S}, [18:3]} 0x8: 16-bit mode, {S,D[18:4]} 0x9: 16-bit mode, {S,D[17:3]} 0xA: 16-bit mode, {S,D[16:2]} 0xB: 16-bit mode, {S,D[15:1]} 0xC: 16-bit mode, {S,D[14:0]} Others: Setting prohibited	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.20 PDSFCRCHn : Sinc Filter Control Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x124 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	SINCRNG[4:0]					SINCDEC[7:0]								—	—	—	—	—	—	—	—	—	—	—	—	CKDIV[3:0]			
Value after reset:	0	0	0	0	0	1	0	1	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CKDIV[3:0]	PDM_CLKn Dividend Ratio to Core Clock 0x0: 2 divisions (default) 0x1: 4 divisions 0x2: 6 divisions : 0xD: 28 divisions 0xE: 30 divisions 0xF: 32 divisions	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W
23:16	SINCDEC[7:0]	Sinc Filter Decimation Ratio M indicates the decimation ratio. Setting less than M = 4 is prohibited. When the value is set, the sinc filter operation is not guaranteed. Set it to satisfy D × M > 12. If it is not satisfied, the operation of other filters is not guaranteed. (D: PDM_CLKn dividend ratio) Decimation ratio M is set as follows. M = SINCDEC + 1 0x00: M = 1 (prohibited) 0x01: M = 2 (prohibited) 0x02: M = 3 (prohibited) 0x03: M = 4 : 0x7C: M = 125 (default) : 0xFF: M = 256	R/W

Bit	Symbol	Function	R/W
28:24	SINCRNG[4:0]	Sinc Filter Output Valid Range When overflow occurs, 0x7FFFF is output. When underflow occurs, 0x80000 is output. S: Signed bit 0x00: {S, [32:14]} 0x01: {S, [31:13]} 0x02: {S, [30:12]} 0x03: {S, [29:11]} 0x04: {S, [28:10]} 0x05: {S, [27:9]} (default) 0x06: {S, [26:8]} 0x07: {S, [25:7]} 0x08: {S, [24:6]} 0x09: {S, [23:5]} 0x0A: {S, [22:4]} 0x0B: {S, [21:3]} 0x0C: {S, [20:2]} 0x0D: {S, [19:1]} 0x0E: {S, [18:0]} 0x0F: {S, [17:0], 0} 0x10: {S, [16:0], 00} 0x11: {S, [15:0], 000} 0x12: {S, [14:0], 0000} 0x13: {S, [13:0], 00000} 0x14: {S, [12:0], 000000} 0x15: {S, [11:0], 0000000} 0x16: {S, [10:0], 00000000} 0x17: {S, [9:0], 000000000} 0x18: {S, [8:0], 0000000000} 0x19: {S, [7:0], 00000000000} 0x1A: {S, [6:0], 000000000000} 0x1B: {S, [5:0], 0000000000000} 0x1C: {S, [4:0], 00000000000000} 0x1D: {S, [3:0], 000000000000000} 0x1E: {S, [2:0], 0000000000000000} 0x1F: {S, [1:0], 00000000000000000} (prohibited)	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

See Table 50.7 for the setting values of SINCDEC[7:0] and SINCRNG[4:0].

50.2.21 PDHFCS0RCHn : High-pass Filter Coefficient s(0) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x128 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	HFS[15:0]															
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	1	1	0	0	0	0	1

Bit	Symbol	Function	R/W
15:0	HFS[15:0]	High-pass Filter Coefficient s(0) 16-bit signed data (Fraction bit [13:0])	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. High-pass filter cut-off frequency = 100 Hz

50.2.22 PDHFCK1RCHn : High-pass Filter Coefficient k(1) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x12C + 0x100 × n

Bit position:	31																15															0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	HFK[15:0]															
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	1	1	0	0	0	0	0	1

Bit	Symbol	Function	R/W
15:0	HFK[15:0]	High-pass Filter Coefficient k(1) 16-bit signed data (Fraction bit [13:0])	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. High-pass filter cut-off frequency = 100 Hz

50.2.23 PDHFCH0RCHn : High-pass Filter Coefficient h(0) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x130 + 0x100 × n

Bit position:	31																15															0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	HFH[15:0]															
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	HFH[15:0]	High-pass Filter Coefficient h(0) 16-bit signed data (Fraction bit [13:0])	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. High-pass filter cut-off frequency = 100 Hz

50.2.24 PDHFCH1RCHn : High-pass Filter Coefficient h(1) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x134 + 0x100 × n

Bit position:	31																15															0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	HFH[15:0]															
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	HFH[15:0]	High-pass Filter Coefficient h(1) 16-bit signed data (Fraction bit [13:0])	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. High-pass filter cut-off frequency = 100 Hz

50.2.25 PDCFCH00RCHn : Compensation Filter Coefficient h(0) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x138 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFH[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	1	0	0	0

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(0) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.26 PDCFCH01RCHn : Compensation Filter Coefficient h(1) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x13C + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFH[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	1

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(1) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.27 PDCFCH02RCHn : Compensation Filter Coefficient h(2) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x140 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFH[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(2) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.28 PDCFCH03RCHn : Compensation Filter Coefficient h(3) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x144 + 0x100 × n

Bit position: 31 … 12 … 0

Bit field: — — — — — — — — — — — — — — — — — — — | CFH[12:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 0 0 1 0 1 0 1 1 0

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(3) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.29 PDCFCH04RCHn : Compensation Filter Coefficient h(4) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x148 + 0x100 × n

Bit position: 31 … 12 … 0

Bit field: — — — — — — — — — — — — — — — — — — — | CFH[12:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 0 1 1 1 0 0

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(4) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.30 PDCFCH05RCHn : Compensation Filter Coefficient h(5) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x14C + 0x100 × n

Bit position: 31 … 12 … 0

Bit field: — — — — — — — — — — — — — — — — — — — | CFH[12:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 1 1 1 0 0 0 0 1

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(5) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.31 PDCFCH06RCHn : Compensation Filter Coefficient h(6) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x150 + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFH[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1	1	1	0	0

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(6) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.32 PDCFCH07RCHn : Compensation Filter Coefficient h(7) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x154 + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFH[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	1	0	1	0	1	1	0

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(7) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.33 PDCFCH08RCHn : Compensation Filter Coefficient h(8) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x158 + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFH[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(8) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.34 PDCFCH09RCHn : Compensation Filter Coefficient h(9) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x15C + 0x100 × n

Bit position:	31																		12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFH[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	1

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(9) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.35 PDCFCH10RCHn : Compensation Filter Coefficient h(10) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x160 + 0x100 × n

Bit position:	31																		12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFH[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	1	0	0	0

Bit	Symbol	Function	R/W
12:0	CFH[12:0]	Compensation Filter Coefficients h(10) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.36 PDLFCH010RCHn : Low-pass Filter Coefficient h0(10) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x164 + 0x100 × n

Bit position:	31																		12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH0[12:0]												
Value after reset[*1]:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	LFH0[12:0]	Low-pass (half-band decimation) Filter Coefficient h0(10) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.37 PDLFCH100RCHn : Low-pass Filter Coefficient h1(0) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x168 + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(0) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.38 PDLFCH101RCHn : Low-pass Filter Coefficient h1(1) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x16C + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(1) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.39 PDLFCH102RCHn : Low-pass Filter Coefficient h1 (2) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x170 + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(2) 13-bit signed data (Fraction bit [10:0])	R/W

Bit	Symbol	Function	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.40 PDLFCH103RCHn : Low-pass Filter Coefficient h1(3) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x174 + 0x100 × n

Bit position: 31 … 12 … 0

Bit field: — — — — — — — — — — — — — — — — — — — | LFH1[12:0]

Value after reset[*1]: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(3) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.41 PDLFCH104RCHn : Low-pass Filter Coefficient h1(4) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x178 + 0x100 × n

Bit position: 31 … 12 … 0

Bit field: — — — — — — — — — — — — — — — — — — — | LFH1[12:0]

Value after reset[*1]: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 0 1 1 1 0 0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(4) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.42 PDLFCH105RCHn : Low-pass Filter Coefficient h1(5) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x17C + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(5) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.43 PDLFCH106RCHn : Low-pass Filter Coefficient h1(6) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x180 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	1	1	0	0	1	1

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(6) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.44 PDLFCH107RCHn : Low-pass Filter Coefficient h1(7) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x184 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1	1	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(7) 13-bit signed data (Fraction bit [10:0])	R/W

Bit	Symbol	Function	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.45 PDLFCH108RCHn : Low-pass Filter Coefficient h1(8) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x188 + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	1	0	1	1	1	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(8) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.46 PDLFCH109RCHn : Low-pass Filter Coefficient h1(9) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x18C + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	0	0	1

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(9) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.47 PDLFCH110RCHn : Low-pass Filter Coefficient h1(10) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x190 + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	0	0	1

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(10) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.48 PDLFCH111RCHn : Low-pass Filter Coefficient h1(11) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x194 + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	1	0	1	1	1	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(11) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.49 PDLFCH112RCHn : Low-pass Filter Coefficient h1(12) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x198 + 0x100 × n

Bit position:	31																			12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1	1	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(12) 13-bit signed data (Fraction bit [10:0])	R/W

Bit	Symbol	Function	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.50 PDLFCH113RCHn : Low-pass Filter Coefficient h1(13) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x19C + 0x100 × n

Bit position:	31																		12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	1	1	0	0	1	1

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(13) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.51 PDLFCH114RCHn : Low-pass Filter Coefficient h1(14) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1A0 + 0x100 × n

Bit position:	31																		12												0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(14) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.52 PDLFCH115RCHn : Low-pass Filter Coefficient h1(15) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1A4 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	1	1	1	0	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(15) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.53 PDLFCH116RCHn : Low-pass Filter Coefficient h1(16) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1A8 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(15) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.54 PDLFCH117RCHn : Low-pass Filter Coefficient h1(17) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1AC + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset*1:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(17) 3-bit signed data (Fraction bit [10:0])	R/W

Bit	Symbol	Function	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.55 PDLFCH118RCHn : Low-pass Filter Coefficient h1(18) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1B0 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset[*1]:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(18) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.56 PDLFCH119RCHn : Low-pass Filter Coefficient h1(19) Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1B4 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LFH1[12:0]												
Value after reset[*1]:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0

Bit	Symbol	Function	R/W
12:0	LFH1[12:0]	Low-pass (half-band decimation) Filter Coefficient h1(19) 13-bit signed data (Fraction bit [10:0])	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Low-pass filter cut-off frequency = 7 kHz

50.2.57 PDSDLTRCHn : Sound Detection Lower Threshold Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1B8 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	SDETL[19:0]																			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
19:0	SDETL[19:0]	Sound Detection Lower Limit This bit sets the lower limit value (20-bit signed data) for sound detection.	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.58 PDSDUTRCHn : Sound Detection Upper Threshold Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1BC + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19–0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	SDETU[19:0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
19:0	SDETU[19:0]	Sound Detection Upper Limit This bit sets the upper limit value (20-bit signed data) for sound detection.	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.59 PDDBCRCHn : Data Buffer Control Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1C0 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	DATRITHR[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	DATRITHR[2:0]	Data Reception Interrupt Threshold 0 0 0: Output interrupt when receiving 1 or more data 0 0 1: Output interrupt when receiving 2 or more data 0 1 0: Output interrupt when receiving 4 or more data 0 1 1: Output interrupt when receiving 8 or more data Others: Output interrupt when receiving 16 or more data	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.60 PDSCTSRCHn : Short Circuit Threshold Setting Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1C4 + 0x100 × n

Bit position:	31			28													16				12												0
Bit field:	—	—	—	SCDH[12:0]														—	—	—	SCDL[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	SCDL[12:0]	Short Circuit Detection Low Continuous Detection Count Set the upper limit for the number of times 0 are input continuously to the PDM_DATAn pin.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
28:16	SCDH[12:0]	Short Circuit Detection High Continuous Detection Count Set the upper limit for the number of times 1 are input continuously to the PDM_DATAn pin.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.61 PDOVLTRCHn : Overvoltage Lower Threshold Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1C8 + 0x100 × n

Bit position:	31												19																		0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	OVDL[19:0]																			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
19:0	OVDL[19:0]	Overvoltage Detection Lower Limit These bits set the lower limit of the active voltage value (20-bit signed data) for overvoltage detection.	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.62 PDOVUTRCHn : Overvoltage Upper Threshold Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1CC + 0x100 × n

Bit position:	31												19																		0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	OVDU[19:0]																			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
19:0	OVDU[19:0]	Overvoltage Detection Upper Limit These bits set the upper limit of the active voltage value (20-bit signed data) for overvoltage detection.	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.63 PDDRCRCHn : Data Read Control Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1E0 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DATRE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DATRE	Data Read Enable Bit 0: Do not allow reading of data from buffer 1: Allow reading of data from buffer	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

50.2.64 PDDCRCHn : Data Clear Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1E4 + 0x100 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DATC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DATC	Data Clear 0: Do nothing 1: Clear data	W
31:1	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

50.2.65 PDDRRCHn : Data Read Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1E8 + 0x100 × n

Bit position:	31												19		0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	DAT[19:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0		

Bit	Symbol	Function	R/W
19:0	DAT[19:0]	Data 20-bit signed sound data or 16-bit signed sound data is read out. For 20-bit mode (PDMDSRCHn.DBIS[3:0] = 0xxx), the signed bit is [19]. For 16-bit mode (PDMDSRCHn.DBIS[3:0] = 1xxx), the signed bits are [19:15]. Writing is invalid.	R
31:20	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

50.2.66 PDDSRCHn : Data Status Register Channel n (n = 0 to 2)

Base address: PDMIF = 0x4025_6000
PDMIF_NS = 0x5025_6000

Offset address: 0x1EC + 0x100 × n

Bit position:	31																								7							0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DATNUM[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	DATNUM[7:0]	Number of Data Stored in Buffer When PDDRCRCHn.DATRE = 0, 0 is read out. When PDDRCRCHn.DATRE = 1, the number of data stored in buffer is read out. For buffer size that is of 32 stages, the maximum value is 31.	R
31:8	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

50.2.67 Mirroring Register

The common registers except for the Version register have mirroring registers in each channel. PDCSR.EDFn is a register which ORs all the error flags of the mirroring registers in each channel.

Use the common registers for synchronous control between channels.

Table 50.4 Mirroring register

Common register		Mirroring register in each channel	
Register name	**Bit name**	**Register name**	**Bit name**
PDCSTRTR	STRTRGn	PDSTRTR	STRTRG
PDCSTPTR	STPTRGn	PDSTPTR	STPTRG
PDCCHGTR	CHGTRGn	PDCHGTR	CHGTRG
PDCICR	ISDEn	PDICR	ISDE
	IDREn		IDRE
	IEDEn		IEDE
PDCSR	STATEn	PDSR	STATE
	SDFn		SDF
	DRFn		DRF
PDCSCR	SDFCn	PDSCR	SDFC
PDCSDCR	SDEn	PDSDCR	SDE
PDCDRCR	DATREn	PDDRCR	DATRE
PDCDCR	DATCn	PDDCR	DATC

50.3 Operation

50.3.1 Clock Specification

The frequency-divided core clock is generated and output to PDM_CLKn.

Table 50.5 shows guaranteed clock frequency and clock mode in PDM-IF. PDM clock (PDMIFCLK) and PCLK are asynchronous.

Table 50.5 Guaranteed clock frequency and clock mode

PDMIFCLK	PCLK	PDM_CLKn		
Frequency (MHz)	Frequency (MHz)	Frequency (MHz)	Supported core clock Division ratio	Setting value (PDSFCR.CKDIV[3:0])
8	1 to 62.5*1	4	2	0x0
		2	4	0x1
		1.33	6	0x2
		1	8	0x3
		0.8	10	0x4
		0.67	12	0x5
		0.57	14	0x6
		0.5	16	0x7
		0.44	18	0x8
		0.4	20	0x9
		0.36	22	0xA
		0.33	24	0xB
		0.31	26	0xC
		0.29	28	0xD
		0.27	30	0xE
		0.25	32	0xF

Note 1. PDM-IF is possible to operate at PCLK below 1 MHz but the guaranteed frequency operation range is 1 to 62.5 MHz.

50.3.1.1 Output Sampling Frequency

PDM-IF output sampling frequency is shown by the following formula:

$F_{out} = F_{clk} / (M \times 2)$

- F_{out}: PDM-IF output sampling frequency
- F_{clk}: PDM_CLKn frequency (See Table 50.5)
- M: Sinc filter decimation ratio (PDSFCRCHn.SINCDEC[7:0] + 1)

50.3.2 Channel Start and Channel Stop

The channel start and the channel stop is controlled by the register of Table 50.6.

Set the channels to be synchronized to 1 at the same time by using the common registers.

Set the channel start according to section 50.4.1. Start Flow.

Set the channel stop according to section 50.4.2. Stop Flow.

Table 50.6 Channel start trigger and channel stop trigger

Category	Register bit name	
	Common register	Channel register
Channel start	PDCSTRTR.STRTRG0	PDSTRTRCH0.STRTRG
	PDCSTRTR.STRTRG1	PDSTRTRCH1.STRTRG
	PDCSTRTR.STRTRG2	PDSTRTRCH2.STRTRG
Channel stop	PDCSTPTR.STPTRG0	PDSTPTRCH0.STPTRG
	PDCSTPTR.STPTRG1	PDSTPTRCH1.STPTRG
	PDCSTPTR.STPTRG2	PDSTPTRCH2.STPTRG

50.3.3 PDM_CLKn Frequency Change

The external microphone transitions the power mode according to input clock frequency. PDM-IF enables to transition the power mode of the microphone by changing the operating frequency of PDM_CLKn which is output to the microphone when operating. The operating frequency of PDM_CLKn can be changed using PDSFCR.CKDIV[3:0] bits. When the frequency is changed, the decimation ratio of the sinc filter must also be changed. Set PDSRCRCHn.SINCDEC[7:0] so that the sampling frequency after the decimation can be 32 kHz*1. Furthermore, when the decimation ratio is changed, the output range of the sinc filter is also changed. Set PDSRCRCHn.SINCRNG[4:0] together in accordance with the output range. See Table 50.7 for details.

After changing the configuration of these registers, software must wait for the settling time of microphone and PDM-IF channel. See section 50.4.4. Low-power Mode Transition Flow and section 50.4.5. Normal Mode Transition Flow for details.

Note 1. In case that the sampling frequency is different from 32 kHz, the characteristics of filters after the sinc filter might change.

50.3.4 Filter

50.3.4.1 Data Clipping and Shifting between Filters

Voice sample data is clipped or shifted between filters. Figure 50.2 shows this operation.

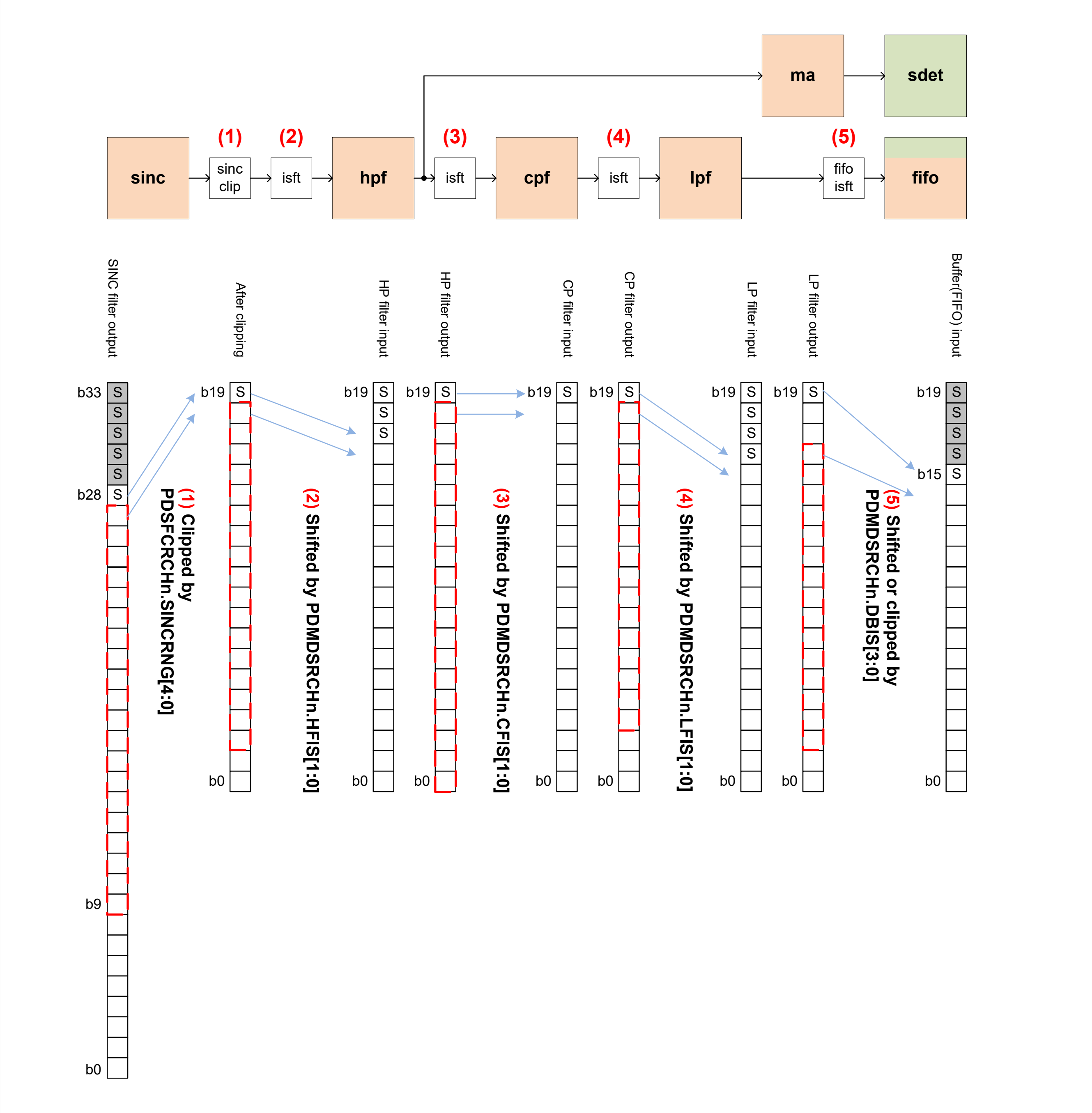


Figure 50.2 Data clipping and shifting between filters

(1) Sinc filter output clip

Table 50.7 lists the available settings of the sinc filter and data clipping.

It is controlled by PDSFCRCHn.SINCRNG[4:0].

It is prohibited to set other than those in Table 50.7.

Table 50.7 Sinc filter output clip (1 of 3)

PDMIFCLK (MHz)	Sinc filter order	PDMDSRCHn.SFMD[2:0]	PDM_CLK dividend ratio	PDSFCRCHn.CKDIV[3:0]	Sinc filter decimation ratio	PDSFCRCHn.SINCDEC[7:0]	Sinc filter output effective bits	Sinc filter output clipping (20 bits)	PDSFCRCHn. SINCRNG[4:0]	Resolution	PDM_CLKn frequency (MHz)	SINC filter output sampling frequency (kHz)	PDM-IF output sampling frequency (kHz)
8	4	0x0	2	0x0	125	0x7C	b28 to b0	b28 to b9	00101	20	4.0	32	16
			4	0x1	62	0x3D	b24 to b0	b24 to b5	01001	20	2.0	32.26	16.13
			6	0x2	42	0x29	b22 to b0	b22 to b3	01011	20	1.33	31.75	15.87
			8	0x3	31	0x1E	b20 to b0	b20 to b1	01101	20	1.0	32.26	16.13
			10	0x4	25	0x18	b19 to b0	b19 to b0	01110	20	0.8	32	16
			12	0x5	21	0x14	b18 to b0	b18 to b0, 0	01111	19	0.67	31.75	15.87
			14	0x6	18	0x11	b17 to b0	b17 to b0, 00	10000	18	0.57	31.75	15.87
			16	0x7	16	0x0F	b17 to b0	b17 to b0, 00	10000	18	0.50	31.25	15.63
			18	0x8	14	0x0D	b16 to b0	b16 to b0, 000	10001	17	0.44	31.75	15.87
			20	0x9	12	0x0B	b15 to b0	b15 to b0, 0000	10010	16	0.40	33.33	16.67
			22	0xA	11	0x0A	b14 to b0	b14 to b0, 00000	10011	15	0.36	33.06	16.53
			24	0xB	10	0x09	b14 to b0	b14 to b0, 00000	10011	15	0.33	33.33	16.67
			26	0xC	9	0x08	b13 to b0	b13 to b0, 000000	10100	14	0.31	34.19	17.09
			28	0xD	9	0x08	b13 to b0	b13 to b0, 000000	10100	14	0.29	31.75	15.87
			30	0xE	8	0x07	b13 to b0	b13 to b0, 000000	10100	14	0.27	33.33	16.67
			32	0xF	8	0x07	b13 to b0	b13 to b0, 000000	10100	14	0.25	31.25	15.63
8	3	0x3	2	0x0	125	0x7C	b21 to b0	b21 to b2	01100	20	4.0	32	16
			4	0x1	62	0x3D	b18 to b0	b18 to b0, 0	01111	19	2.0	32.26	16.13
			6	0x2	42	0x29	b17 to b0	b17 to b0, 00	10000	18	1.33	31.75	15.87
			8	0x3	31	0x1E	b15 to b0	b15 to b0, 0000	10010	16	1.0	32.26	16.13
			10	0x4	25	0x18	b14 to b0	b14 to b0, 00000	10011	15	0.8	32	16
			12	0x5	21	0x14	b14 to b0	b14 to b0, 00000	10011	15	0.67	31.75	15.87
			14	0x6	18	0x11	b13 to b0	b13 to b0, 000000	10100	14	0.57	31.75	15.87
			16	0x7	16	0x0F	b13 to b0	b13 to b0, 000000	10100	14	0.50	31.25	15.63
			18	0x8	14	0x0D	b12 to b0	b12 to b0, 0000000	10101	13	0.44	31.75	15.87
			20	0x9	12	0x0B	b11 to b0	b11 to b0, 00000000	10110	12	0.40	33.33	16.67
			22	0xA	11	0x0A	b11 to b0	b11 to b0, 00000000	10110	12	0.36	33.06	16.53
			24	0xB	10	0x09	b10 to b0	b10 to b0, 000000000	10111	11	0.33	33.33	16.67
			26	0xC	9	0x08	b10 to b0	b10 to b0, 000000000	10111	11	0.31	34.19	17.09
			28	0xD	9	0x08	b10 to b0	b10 to b0, 000000000	10111	11	0.29	31.75	15.87
			30	0xE	8	0x07	b10 to b0	b10 to b0, 000000000	10111	11	0.27	33.33	16.67
			32	0xF	8	0x07	b10 to b0	b10 to b0, 000000000	10111	11	0.25	31.25	15.63

Table 50.7 Sinc filter output clip (2 of 3)

PDMIFCLK (MHz)	Sinc filter order	PDMDSRCHn.SFMD[2:0]	PDM_CLK dividend ratio	PDSFCRCHn.CKDIV[3:0]	Sinc filter decimation ratio	PDSFCRCHn.SINCDEC[7:0]	Sinc filter output effective bits	Sinc filter output clipping (20 bits)	PDSFCRCHn. SINCRNG[4:0]	Resolution	PDM_CLKn frequency (MHz)	SINC filter output sampling frequency (kHz)	PDM-IF output sampling frequency (kHz)
8	2	0x2	2	0x0	125	0x7C	b14 to b0	b14 to b0, 00000	10011	15	4.0	32	16
			4	0x1	62	0x3D	b12 to b0	b12 to b0, 0000000	10101	13	2.0	32.26	16.13
			6	0x2	42	0x29	b11 to b0	b11 to b0, 00000000	10110	12	1.33	31.75	15.87
			8	0x3	31	0x1E	b10 to b0	b10 to b0, 000000000	10111	11	1.0	32.26	16.13
			10	0x4	25	0x18	b10 to b0	b10 to b0, 000000000	10111	11	0.8	32	16
			12	0x5	21	0x14	b9 to b0	b9 to b0, 0000000000	11000	10	0.67	31.75	15.87
			14	0x6	18	0x11	b9 to b0	b9 to b0, 0000000000	11000	10	0.57	31.75	15.87
			16	0x7	16	0x0F	b9 to b0	b9 to b0, 0000000000	11000	10	0.50	31.25	15.63
			18	0x8	14	0x0D	b8 to b0	b8 to b0, 00000000000	11001	9	0.44	31.75	15.87
			20	0x9	12	0x0B	b8 to b0	b8 to b0, 00000000000	11001	9	0.40	33.33	16.67
			22	0xA	11	0x0A	b7 to b0	b7 to b0, 000000000000	11010	8	0.36	33.06	16.53
			24	0xB	10	0x09	b7 to b0	b7 to b0, 000000000000	11010	8	0.33	33.33	16.67
			26	0xC	9	0x08	b7 to b0	b7 to b0, 000000000000	11010	8	0.31	34.19	17.09
			28	0xD	9	0x08	b7 to b0	b7 to b0, 000000000000	11010	8	0.29	31.75	15.87
			30	0xE	8	0x07	b7 to b0	b7 to b0, 000000000000	11010	8	0.27	33.33	16.67
			32	0xF	8	0x07	b7 to b0	b7 to b0, 000000000000	11010	8	0.25	31.25	15.63

Table 50.7 Sinc filter output clip (3 of 3)

PDMIFCLK (MHz)	Sinc filter order	PDMDSRCHn.SFMD[2:0]	PDM_CLK dividend ratio	PDSFCRCHn.CKDIV[3:0]	Sinc filter decimation ratio	PDSFCRCHn.SINCDEC[7:0]	Sinc filter output effective bits	Sinc filter output clipping (20 bits)	PDSFCRCHn. SINCRNG[4:0]	Resolution	PDM_CLKn frequency (MHz)	SINC filter output sampling frequency (kHz)	PDM-IF output sampling frequency (kHz)
8	1	0x1	2	0x0	125	0x7C	b7 to b0	b7 to b0, 000000000000	11010	8	4.0	32	16
			4	0x1	62	0x3D	b6 to b0	b6 to b0, 0000000000000	11011	7	2.0	32.26	16.13
			6	0x2	42	0x29	b6 to b0	b6 to b0, 0000000000000	11011	7	1.33	31.75	15.87
			8	0x3	31	0x1E	b5 to b0	b5 to b0, 00000000000000	11100	6	1.0	32.26	16.13
			10	0x4	25	0x18	b5 to b0	b5 to b0, 00000000000000	11100	6	0.8	32	16
			12	0x5	21	0x14	b5 to b0	b5 to b0, 00000000000000	11100	6	0.67	31.75	15.87
			14	0x6	18	0x11	b5 to b0	b5 to b0, 00000000000000	11100	6	0.57	31.75	15.87
			16	0x7	16	0x0F	b5 to b0	b5 to b0, 00000000000000	11100	6	0.50	31.25	15.63
			18	0x8	14	0x0D	b4 to b0	b4 to b0, 000000000000000	11101	5	0.44	31.75	15.87
			20	0x9	12	0x0B	b4 to b0	b4 to b0, 000000000000000	11101	5	0.40	33.33	16.67
			22	0xA	11	0x0A	b4 to b0	b4 to b0, 000000000000000	11101	5	0.36	33.06	16.53
			24	0xB	10	0x09	b4 to b0	b4 to b0, 000000000000000	11101	5	0.33	33.33	16.67
			26	0xC	9	0x08	b4 to b0	b4 to b0, 000000000000000	11101	5	0.31	34.19	17.09
			28	0xD	9	0x08	b4 to b0	b4 to b0, 000000000000000	11101	5	0.29	31.75	15.87
			30	0xE	8	0x07	b4 to b0	b4 to b0, 000000000000000	11101	5	0.27	33.33	16.67
			32	0xF	8	0x07	b4 to b0	b4 to b0, 000000000000000	11101	5	0.25	31.25	15.63

An overflow or an underflow never occurs as long as the combinations which are shown in Table 50.7 are set to PDMDSRCHn.SFMD[2:0], PDSFCRCHn.CKDIV[3:0], PDSFCRCHn.SINCDEC[7:0] and PDSFCRCHn.SINCRNG[4:0]. However, when other combinations are set to them, the overflow or the underflow might occur. When the overflow occurs, 0x7FFFF is output. When the underflow occurs, 0x80000 is output. Use the overvoltage detection function if you want to know if the overflow or the underflow occurs.

(2) High-pass filter input shift

The clipped sinc filter output is right-shifted and input to the high-pass filter.

It is controlled by PDMDSRCHn.HFIS[1:0].

Table 50.8 High-pass filter input shift (1 of 2)

PDMDSR.HFIS[1:0]	Function	Shifted data[*1]
0x0	No shift	S, b18-b0
0x1	1-bit right-shift	S, S, b18-b1

Table 50.8 High-pass filter input shift (2 of 2)

PDMDSR.HFIS[1:0]	Function	Shifted data*1
0x2	2-bit right-shift	S, S, S, b18-b2
0x3	3-bit right-shift	S, S, S, S, b18-b3

Note 1. S: Signed bit (= bit 19)

(3) Compensation filter input shift

The high-pass filter output is right-shifted and input to the compensation filter.

It is controlled by PDMDSRCHn.CFIS[1:0].

Table 50.9 Compensation filter input shift

PDMDSR.CFIS[1:0]	Function	Shifted data*1
0x0	No shift	S, b18-b0
0x1	1-bit right-shift	S, S, b18-b1
0x2	2-bit right-shift	S, S, S, b18-b2
0x3	3-bit right-shift	S, S, S, S, b18-b3

Note 1. S: Signed bit (= bit 19)

(4) Low-pass filter input shift

The compensation filter output is right-shifted and input to the low-pass (half-band decimation) filter.

It is controlled by PDMDSRCHn.LFIS[1:0].

Table 50.10 Low-pass filter input shift

PDMDSR.LFIS[1:0]	Function	Shifted data*1
0x0	No shift	S, b18-b0
0x1	1-bit right-shift	S, S, b18-b1
0x2	2-bit right-shift	S, S, S, b18-b2
0x3	3-bit right-shift	S, S, S, S, b18-b3

Note 1. S: Signed bit (= bit 19)

(5) Buffer input shift or clip

The compensation filter output is right-shifted, or 16-bit data is clipped from the compensation filter output. The shifted or clipped data is then input to buffer.

It is controlled by PDMDSRCHn.DBIS[1:0].

Table 50.11 Buffer input shift or clip (1 of 2)

PDMDSRCHn.DBIS[3:0]	Function	Valid data width	Shifted or Clipped data*1
0000	No shift	20-bit	S, b18-b0
0001	1-bit right-shift	20-bit	S, S, b18-b1
0010	2-bit right-shift	20-bit	S, S, S, b18-b2
0011	3-bit right-shift	20-bit	S, S, S, S, b18-b3
1000	Signed bit and 18-4 is clipped	16-bit	S, b18-b4
1001	Signed bit and 17-3 is clipped*2	16-bit	S, b17-b3
1010	Signed bit and 16-2 is clipped*2	16-bit	S, b16-b2

Table 50.11 Buffer input shift or clip (2 of 2)

PDMDSRCHn.DBIS[3:0]	Function	Valid data width	Shifted or Clipped data*1
1011	Signed bit and 15-1 is clipped*2	16-bit	S, b15-b1
1100	Signed bit and 14-0 is clipped*2	16-bit	S, b14-b0

Note 1. S: Signed bit (= bit 19)
Note 2. When an overflow occurs, 0x7FFF is input to the buffer.
When an underflow occurs, 0x8000 is input to the buffer.

50.3.4.2 Sinc Filter

The operating clock of the sinc filter is PDMIFCLK. Sinc filter is operated using the 1-bit stream data latched with the rise-edge or fall-edge of PDM_CLKn. For details, see section 50.3.5. Data Input Control for Stereo Sound.

Figure 50.3 shows a sinc filter block diagram.

The differentiation stage is operated by decimation clock (the frequency is 1/M of PDM_CLKn). M is the decimation ratio and is set in the PDSFCRCHn.SINCDEC[7:0] bits. The filter result is output in register every decimation clock.

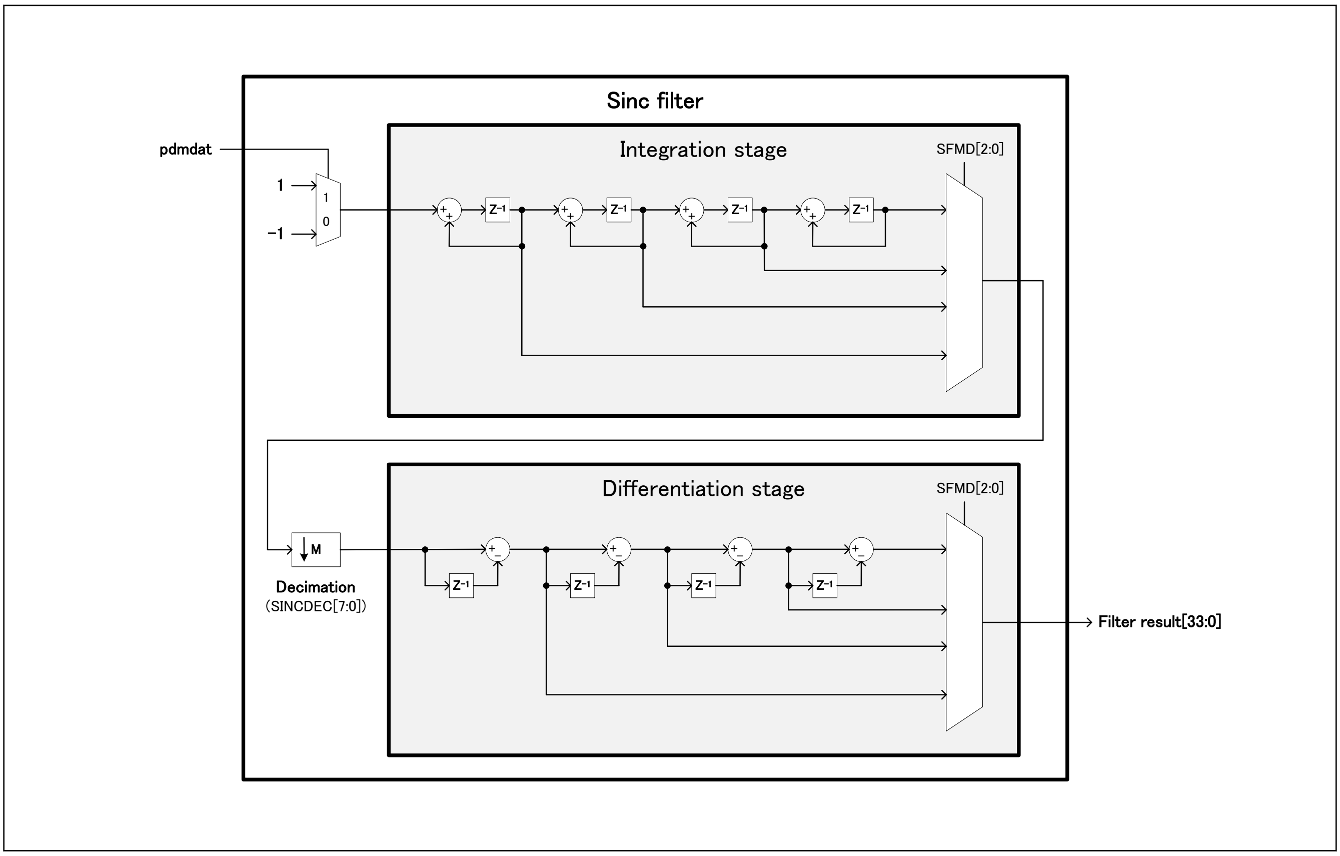


Figure 50.3 Sinc filter block diagram

50.3.4.3 High-pass Filter

The operating clock of the high-pass filter is PDMIFCLK. High-pass filter is operated using the result of the sinc filter.

Figure 50.4 shows a high-pass filter block diagram.

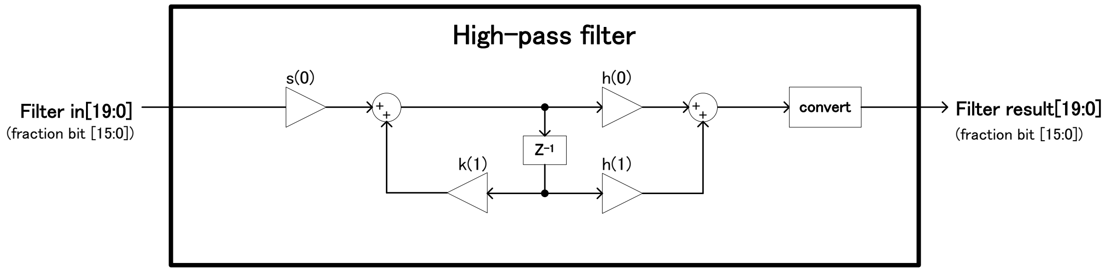


Figure 50.4 High-pass filter block diagram

Table 50.12 shows a high-pass filter specification.

Table 50.12 High-pass filter specification

Function		Specification
Input	Bit width	Signed 20-bit (fraction 16-bit)
Output	Bit width	Signed 20-bit (fraction 16-bit)
Coefficient	Bit width	Signed 16-bit (fraction 14-bit) For details, see section 50.2.21. PDHFCS0RCHn : High-pass Filter Coefficient s(0) Register Channel n (n = 0 to 2), section 50.2.22. PDHFCK1RCHn : High-pass Filter Coefficient k(1) Register Channel n (n = 0 to 2), and section 50.2.23. PDHFCH0RCHn : High-pass Filter Coefficient h(0) Register Channel n (n = 0 to 2).
Delay Z	Bit width	Signed 32-bit (fraction 18-bit)
Multiplier	Input A bit width	Signed 32-bit (fraction 18-bit)
	Input B bit width	Signed 16-bit (fraction 14-bit)
	Output bit width	Signed 47-bit (fraction 32-bit)
Adder	Input A bit width	Signed 32-bit (fraction 18-bit)
	Input B bit width	Signed 32-bit (fraction 18-bit)
	Output bit width	Signed 33-bit (fraction 18-bit)
Output conversion	Input bit width	Signed 32-bit (fraction 18-bit)
	Output bit width	Signed 20-bit (fraction 16-bit)
Excessive fraction bits processing		Excessive fraction bits are rounded down at functional unit input or final output

50.3.4.4 Compensation Filter

The operating clock of the compensation filter is PDMIFCLK. Compensation filter is operated using the result of the high-pass filter.

Figure 50.5 shows a compensation filter block diagram.

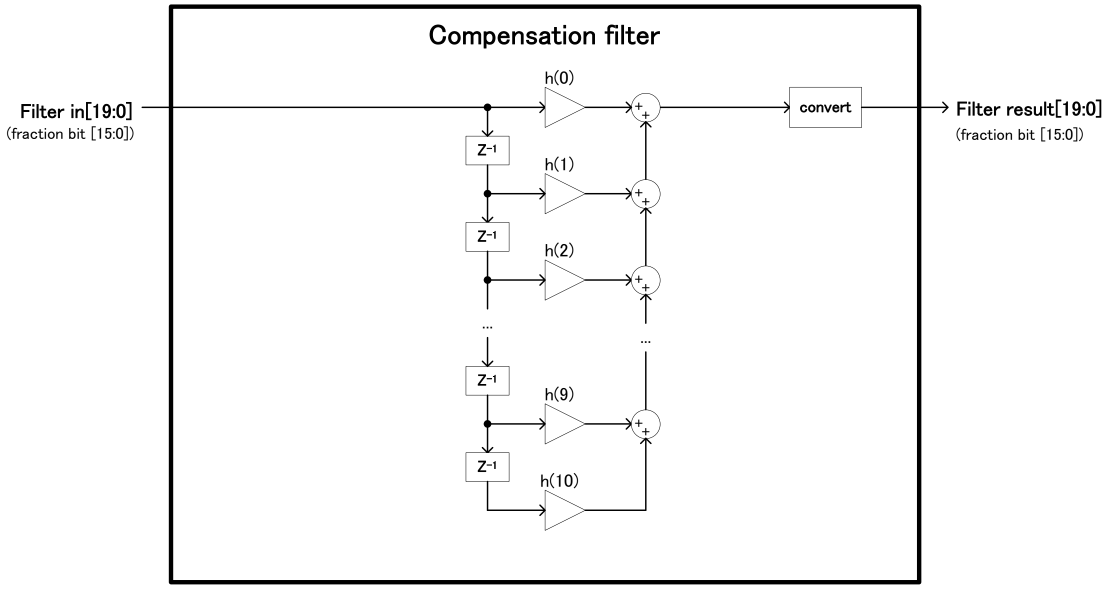


Figure 50.5 Compensation filter block diagram

Table 50.13 shows a compensation filter specification.

Table 50.13 Compensation filter specification

Function		Specification
Input	Bit width	Signed 20-bit (fraction 16-bit)
Output	Bit width	Signed 20-bit (fraction 16-bit)
Coefficient	Bit width	Signed 13-bit (fraction 11-bit) For details, see section 50.2.25. PDCFCH00RCHn : Compensation Filter Coefficient h(0) Register Channel n (n = 0 to 2).
Delay Z	Bit width	Signed 20-bit (fraction 16-bit)
Multiplier	Input A bit width	Signed 20-bit (fraction 16-bit)
	Input B bit width	Signed 13-bit (fraction 11-bit)
	Output bit width	Signed 32-bit (fraction 27-bit)
Adder	Input A bit width	Signed 22-bit (fraction 18-bit)
	Input B bit width	Signed 22-bit (fraction 18-bit)
	Output bit width	Signed 23-bit (fraction 18-bit)
Output conversion	Input bit width	Signed 22-bit (fraction 18-bit)
	Output bit width	Signed 20-bit (fraction 16-bit)
Excessive fraction bits processing		Excessive fraction bits are rounded down at functional unit input or final output

50.3.4.5 Low-pass (Half-band Decimation) Filter

The operating clock of the low-pass filter is PDMIFCLK. Low-pass filter is operated using the result of the compensation filter. Output data is decimated to a half of input data. As a result, output data frequency reduces to half of input frequency and the output data frequency is equal to the sample rate of data stored in FIFO.

Figure 50.6 shows a low-pass filter block diagram.

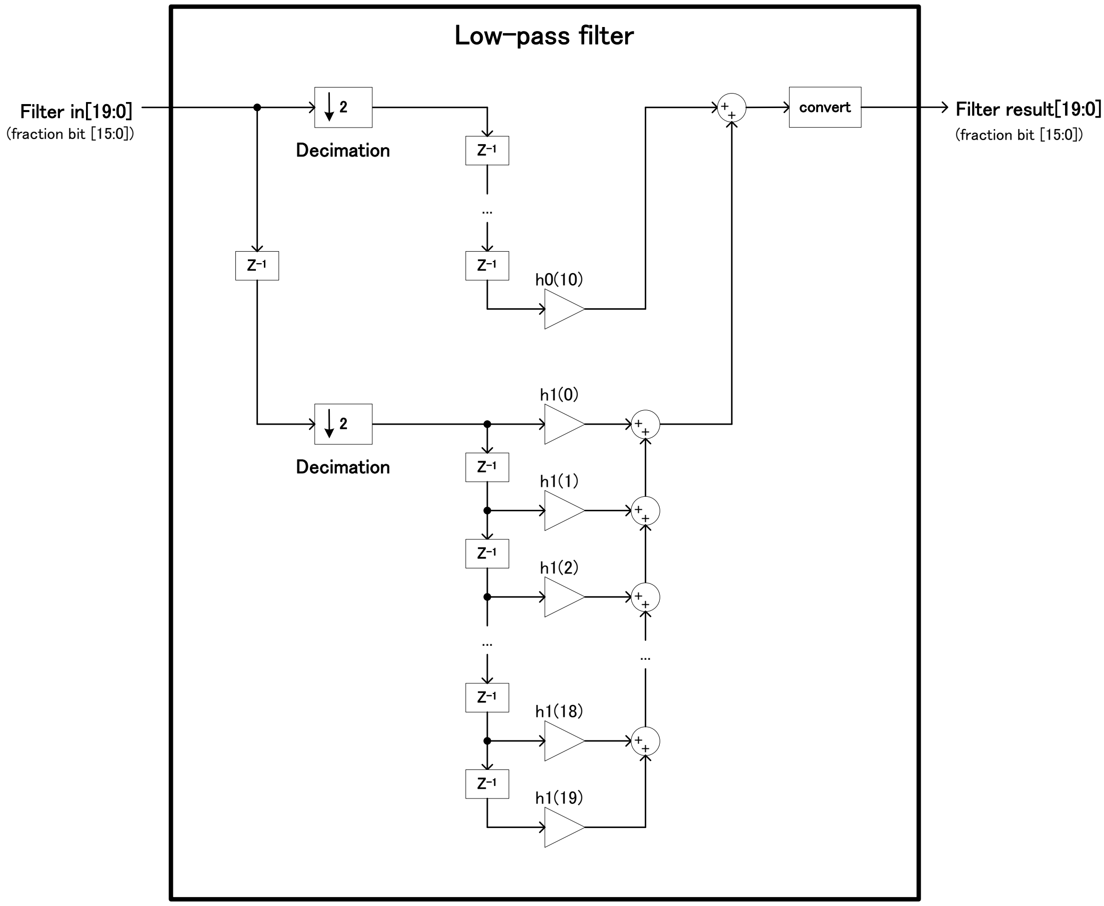


Figure 50.6 Low-pass filter block diagram

Table 50.14 shows a low-pass filter specification.

Table 50.14 Low-pass filter specification

Function		Specification
Input	Bit width	Signed 20-bit (fraction 16-bit)
Output	Bit width	Signed 20-bit (fraction 16-bit)
Coefficient	Bit width	Signed 13-bit (fraction 11-bit) For details, see section 50.2.25. PDCFCH00RCHn : Compensation Filter Coefficient h(0) Register Channel n (n = 0 to 2).
Delay Z	Bit width	Signed 20-bit (fraction 16-bit)
Multiplier	Input A bit width	Signed 20-bit (fraction 16-bit)
	Input B bit width	Signed 13-bit (fraction 11-bit)
	Output bit width	Signed 32-bit (fraction 27-bit)
Adder	Input A bit width	Signed 22-bit (fraction 18-bit)
	Input B bit width	Signed 22-bit (fraction 18-bit)
	Output bit width	Signed 23-bit (fraction 18-bit)
Output conversion	Input bit width	Signed 22-bit (fraction 18-bit)
	Output bit width	Signed 20-bit (fraction 16-bit)
Excessive fraction bits processing		Excessive fraction bits are rounded down at functional unit input or final output

50.3.4.6 Moving Average Filter

The operating clock of the moving average filter is PDMIFCLK. The moving average filter is operated using the result of the high-pass filter. The order of the moving average filter can be changed by the setting of the PDMDSRCHn.SDMAMD[1:0] bits.

Figure 50.7 shows a moving average filter block diagram.

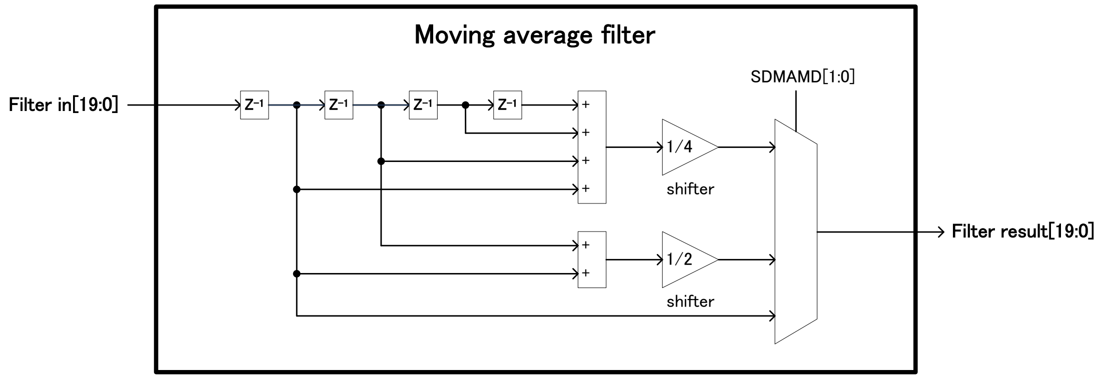


Figure 50.7 Moving average filter block diagram

Table 50.15 shows a moving average filter specification.

Table 50.15 Moving average filter specification

Function		Specification
Input	Bit width	Signed 20-bit (fraction 16-bit)
Output	Bit width	Signed 20-bit (fraction 16-bit)
Delay Z	Bit width	Signed 20-bit (fraction 16-bit)
Excessive fraction bits processing		Excessive fraction bits are rounded down at shifter

50.3.4.7 Settling Time of Channel Activation/Setting Change

Settling time of channel activation is the time after a start trigger PDCSTRTR.STRTRGn (or PDSTRTRCHn.STRTRG) is set before the filter result is stable and output. Settling time of channel setting change is the time after a change trigger PDCCHGTR.CHGTRGn (or PDCHGTRCHn.CHGTRG) is set before the filter result is stable and output. Figure 50.8 shows the settling time measurement point.

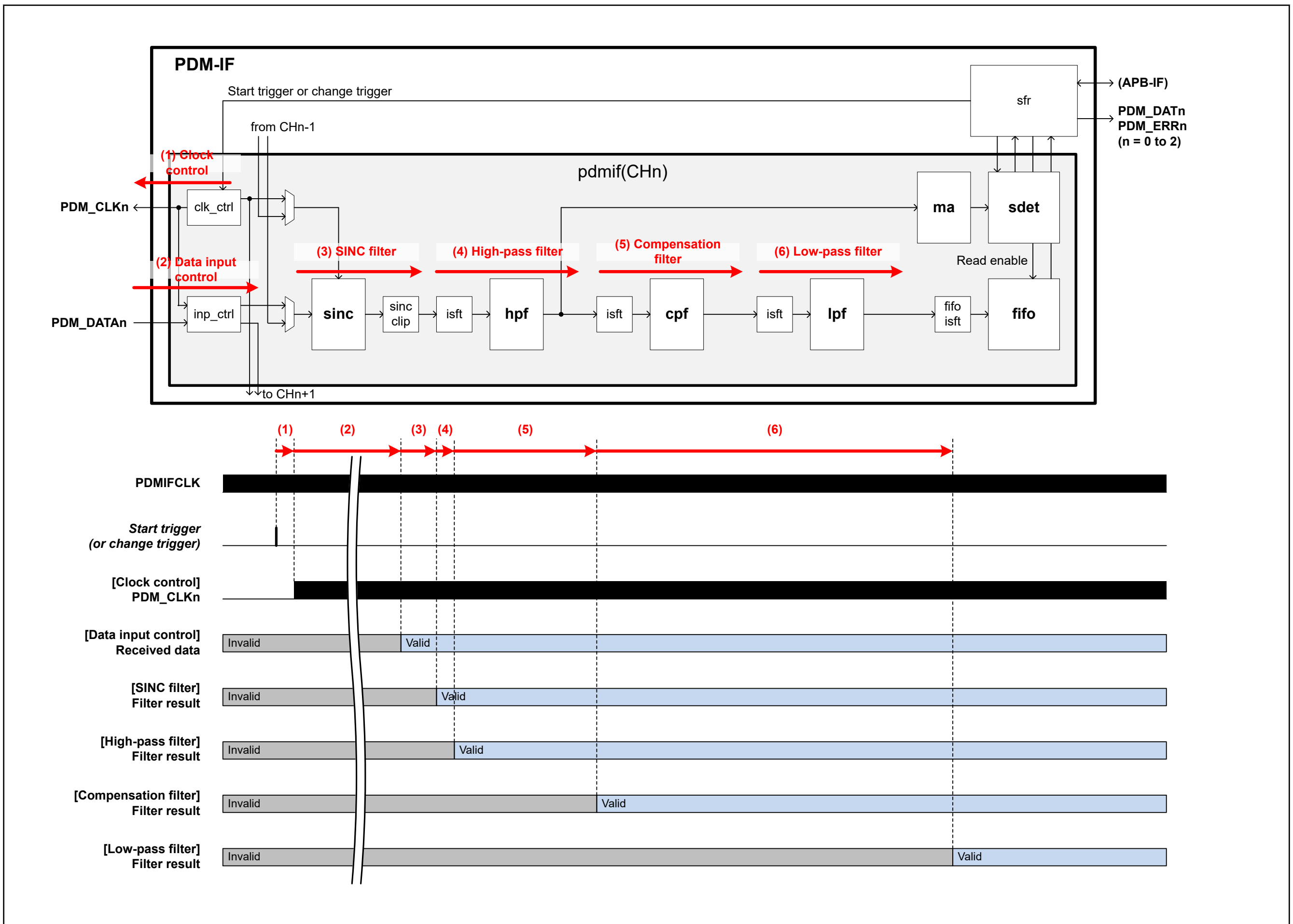


Figure 50.8 Settling time of channel activation

Settling time is calculated as Table 50.16.

Table 50.16 Settling time of channel activation/setting change

No	Function	Settling time	
		Channel activation	Channel setting change
(1)	Clock control	(4 + (D / 2)) × Cp	(3 + Do + (D / 2)) × Cp
(2)	Data input control	(D / 2) × Cp + (***Stable time for microphone***[*1])	
(3)	Sinc filter	(K × D × M) × Cp	
(4)	High-pass filter	(D × M + 5) × Cp	
(5)	Compensation filter	(10 × D × M + 12) × Cp	
(6)	Low-pass filter	(38 × D × M + 22) × Cp	

Note 1. Wakeup time from low power mode, switching time. For details, see datasheet of microphone.

Cp: PDMIFCLK period

D: PDM_CLKn’s dividend ratio

Do: PDM_CLKn’s dividend ratio (old setting)

K: Sinc filter order

M: Sinc filter decimation ratio

50.3.5 Data Input Control for Stereo Sound

The microphone which supports the stereo sound outputs the right sound synchronized with the rise-edge of PDM_CLKn and the left sound synchronized with the fall-edge of PDM_CLKn.

PDM-IF can process the right sound and the left sound using different channels.

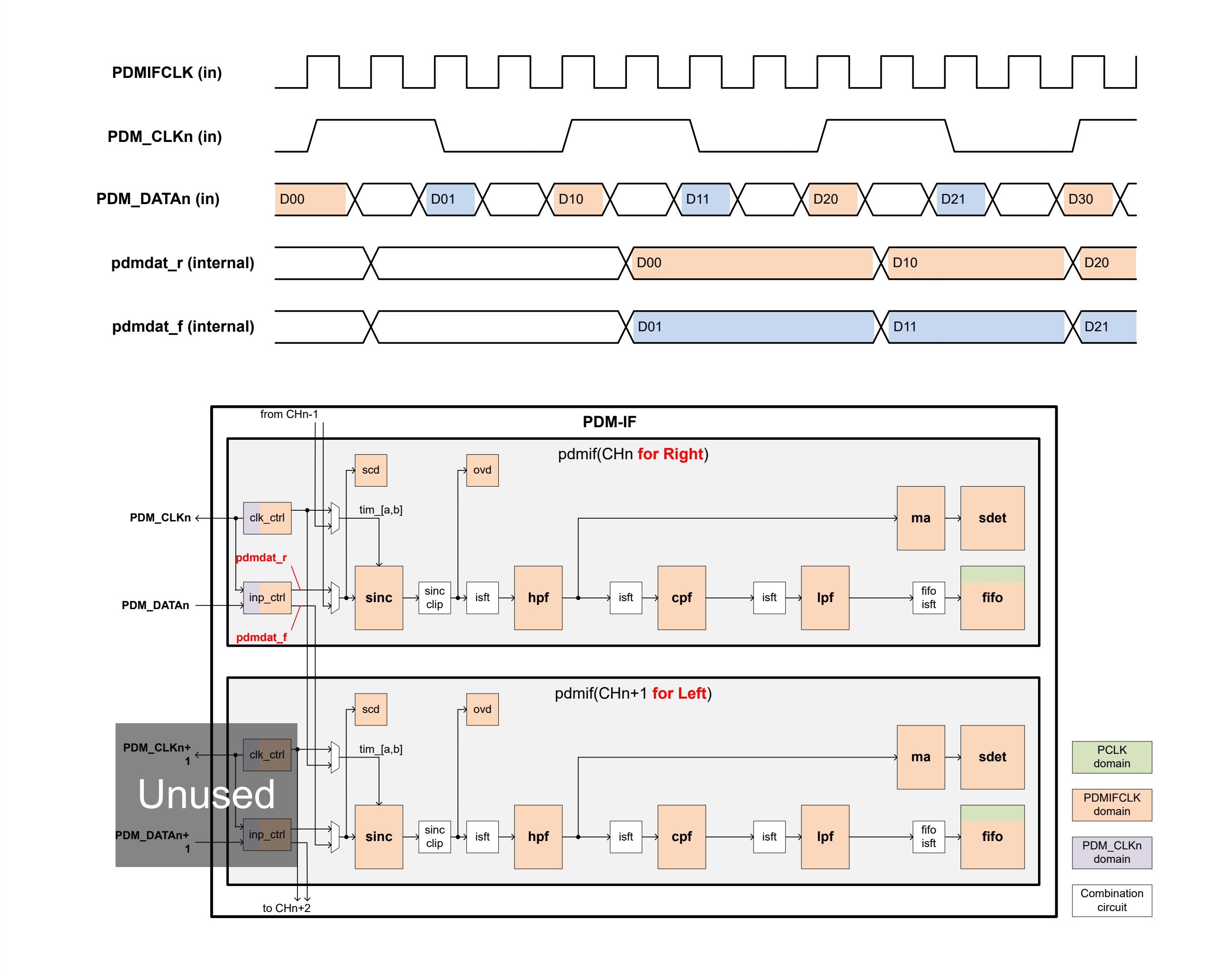


Figure 50.9 Data input control for stereo sound

Table 50.17 shows channel combinations of stereo sound processing.

Table 50.17 Channel combinations of stereo sound processing

Input	Data processing latched at rise-edge (right sound)	Data processing latched at fall-edge (left sound)
PDM_CLK0	Channel 0	Channel 1
PDM_DATA0		
PDM_CLK1	Channel 1	Channel 0
PDM_DATA1		

Table 50.18 shows an example of register settings for stereo sound processing using channel 0 and channel 1.

Table 50.18 Register settings for stereo sound processing (1 of 2)

Register	Bit	Channel 0 setting	Channel 1 setting
PDICRCHn	All bits	Channel unique settings	Channel unique settings
PDSDCRCHn	All bits	Channel unique settings	Channel unique settings

Table 50.18 Register settings for stereo sound processing (2 of 2)

Register	Bit	Channel 0 setting	Channel 1 setting
PDMDSRCHn	INPSEL	0 (use channel 0 data synchronized with rise-edge)	1 (use channel 0 data synchronized with fall-edge)
	SFMD[2:0]	Same settings for channel 0 and channel 1 (mandatory)	
	HFIS[1:0]	Channel unique settings	Channel unique settings
	HFMD	Same settings for channel 0 and channel 1 (mandatory)	
	CFIS[1:0]	Channel unique settings	Channel unique settings
	CFMD	Same settings for channel 0 and channel 1 (mandatory)	
	LFIS[1:0]	Channel unique settings	Channel unique settings
	LFMD	Same settings for channel 0 and channel 1 (mandatory)	
	SDHFMD	Same settings for channel 0 and channel 1 (mandatory)	
	SDMAMD[1:0]	Same settings for channel 0 and channel 1 (mandatory)	
	DBIS[3:0]	Channel unique settings	Channel unique settings
PDSFCRCHn	CKDIV[3:0]	Setting of the clock which is supplied to the microphone connected to channel 0	Unused
	SINCDEC[7:0]	Same settings for channel 0 and channel 1 (mandatory)	
	SINCRNG[4:0]	Same settings for channel 0 and channel 1 (mandatory)	
PDHFCS0RCHn	All bits	Channel unique settings	Channel unique settings
PDHFCK1RCHn	All bits	Channel unique settings	Channel unique settings
PDHFCH0RCHn	All bits	Channel unique settings	Channel unique settings
PDHFCH1RCHn	All bits	Channel unique settings	Channel unique settings
PDCFCHmRCHn	All bits	Channel unique settings	Channel unique settings
PDLFCH0mRCHn	All bits	Channel unique settings	Channel unique settings
PDLFCH1mRCHn	All bits	Channel unique settings	Channel unique settings
PDSDLTRCHn	All bits	Channel unique settings	Channel unique settings
PDSDUTRCHn	All bits	Channel unique settings	Channel unique settings
PDDBCRCHn	All bits	Same settings for channel 0 and channel 1 (mandatory)	
PDSCTSRCHn	All bits	Channel unique settings	Channel unique settings
PDOVLTRCHn	All bits	Channel unique settings	Channel unique settings
PDOVUTRCHn	All bits	Channel unique settings	Channel unique settings

50.3.6 Sound Detection

The operating clock of the sound detection function is PDMIFCLK. When PDSDCRCHn.SDE = 1, the function determines the sound detection using the moving average filter results.

A sound detection flag (PDSR.SDF) is set to 1 when the result is above the upper limit PDSDUTRCHn.SDETU[19:0]. The sound detection flag is also set to 1 when the result is below the lower limit PDSDLTRCHn.SDETL[19:0]. The sound detection flag is cleared to 0 when 1 is written to PDSCRCHn.SDFC.

Figure 50.10 shows the sound detection function.

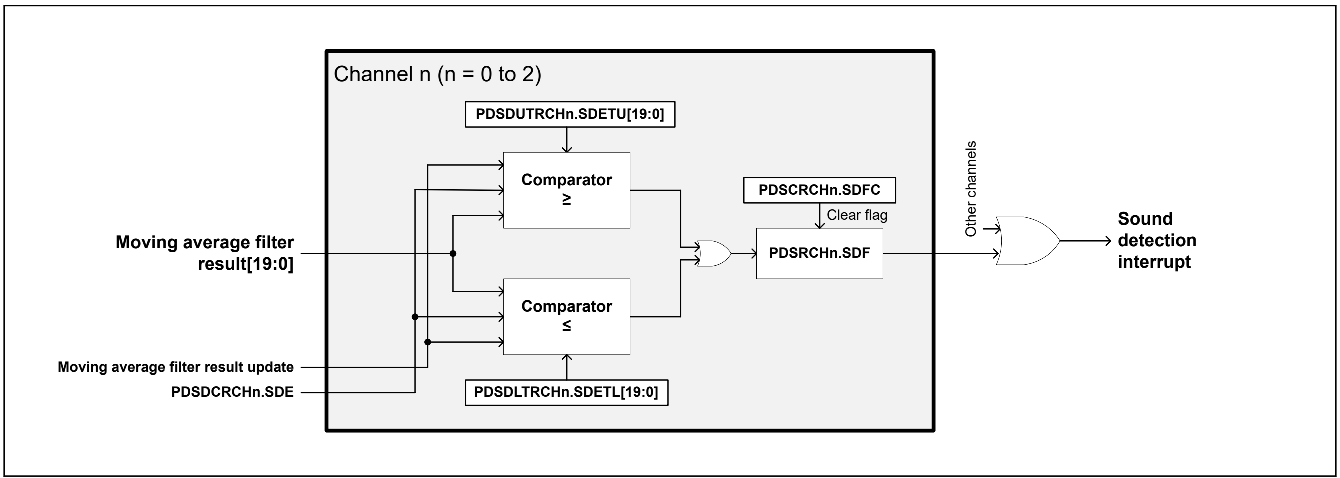


Figure 50.10 Sound detection function

50.3.7 Data Buffer

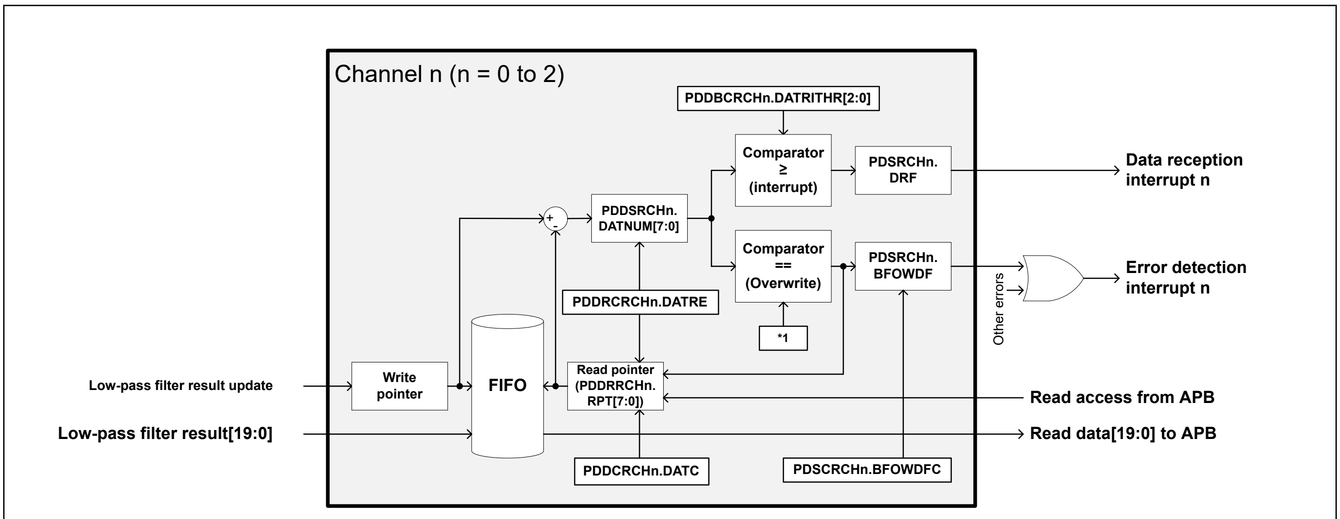
Data buffer writes sound data to FIFO synchronizing with PDMIFCLK and reads sound data from FIFO synchronizing with PCLK. Data buffer stores sound data in FIFO until it is read out.

While PDDRCRCHn.DATRE = 0 (hereinafter, DATRE), PDDSR.DATNUM[7:0] (hereinafter, DATNUM[7:0]) shows 0 and PDM-IF does not allow sound data to be read. After DATRE is changed from 0 to 1, DATNUM[7:0] shows the buffer size minus 2 and PDM-IF allows sound data to be read. While DATRE = 1, data buffer increments DATNUM[7:0] when data is written and decrements DATNUM[7:0] when data is read.

When DATNUM[7:0] is higher than or equal to a threshold which is configured by PDDBCRCHn.DATRITHR[2:0], a data reception flag PDSRCHn.DRF (hereinafter, DRF) is set to 1. When DATNUM[7:0] is lower than the threshold by reading data for instance, a data reception flag DRF is cleared to 0.

When DATNUM[7:0] is the buffer size minus 2 and data is written without reading, data buffer overwrites the oldest data with a new one and a buffer overwriting detection flag PDSRCHn.BFOWDF (hereinafter, BFOWDF) is set to 1. DATNUM[7:0] is then set to the buffer size minus 2 again after overwriting. However, data buffer overwrites but BFOWDF is not set after DATRE changes from 0 to 1 to before the first data is read.

Figure 50.11 shows the data buffer function.



Note 1. Buffer size - 1 (when buffer size is of 32 stages, this value is 31).

Figure 50.11 Data buffer function

50.3.8 Short-circuit Detection

When short-circuit detection enable (PDSDCRCHn.SCDE) is valid, the short-circuit detection can operate.

A dedicated 13-bit counter counts consecutive 0s or 1s input from pdmdat which is output by data input control block of channel n or channel n + 1.

When the number of consecutive 0s exceeds the value set in the PDSCTSRCHn.SCDL[12:0] register or the number of consecutive 1s exceeds the value set in the PDSCTSRCHn.SCDH[12:0] register, a short-circuit detection interrupt request is generated.

Figure 50.12 shows the configuration for the short-circuit detection function.

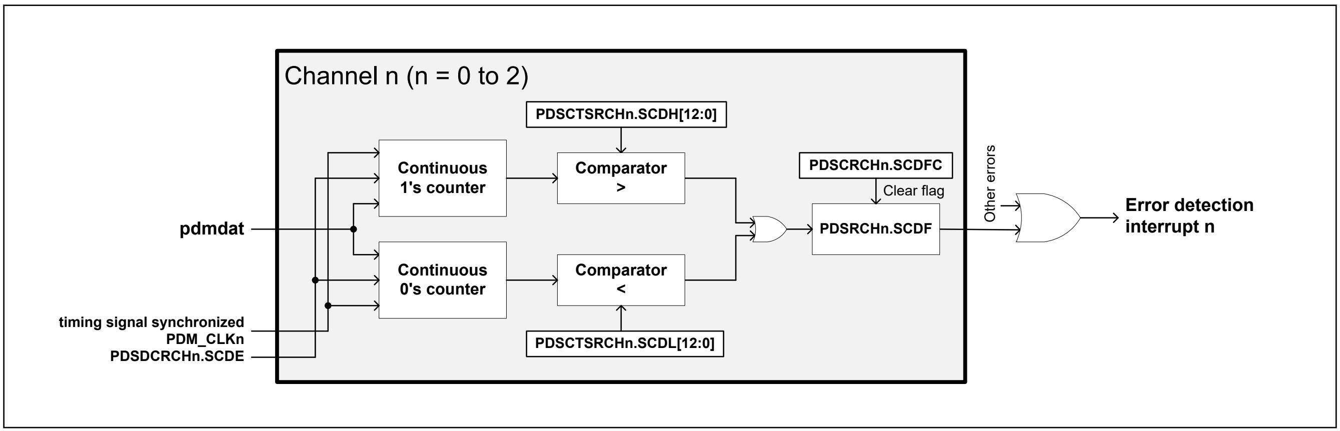


Figure 50.12 Short-circuit detection function

50.3.9 Overvoltage Detection

When the overvoltage upper limit detection enable (PDSDCRCHn.OVUDE) is valid, the overvoltage upper limit detection can operate. When the overvoltage lower limit detection enable (PDSDCRCHn.OVLDE) is valid, the overvoltage lower limit detection can operate.

The sound data is clipped as a 20-bit data from 34-bit data output from sinc filter by setting the PDSFCR.SINCRNG[4:0] bits. See Table 50.7 for detail.

An overvoltage upper limit detection flag is set to 1 when the clipped sound data is higher than PDOVUTRCHn.OVDU[19:0].

An overvoltage lower limit detection flag is set to 1 when the clipped sound data is lower than PDOVUTRCHn.OVDL[19:0].

Figure 50.13 shows the overvoltage detection function.

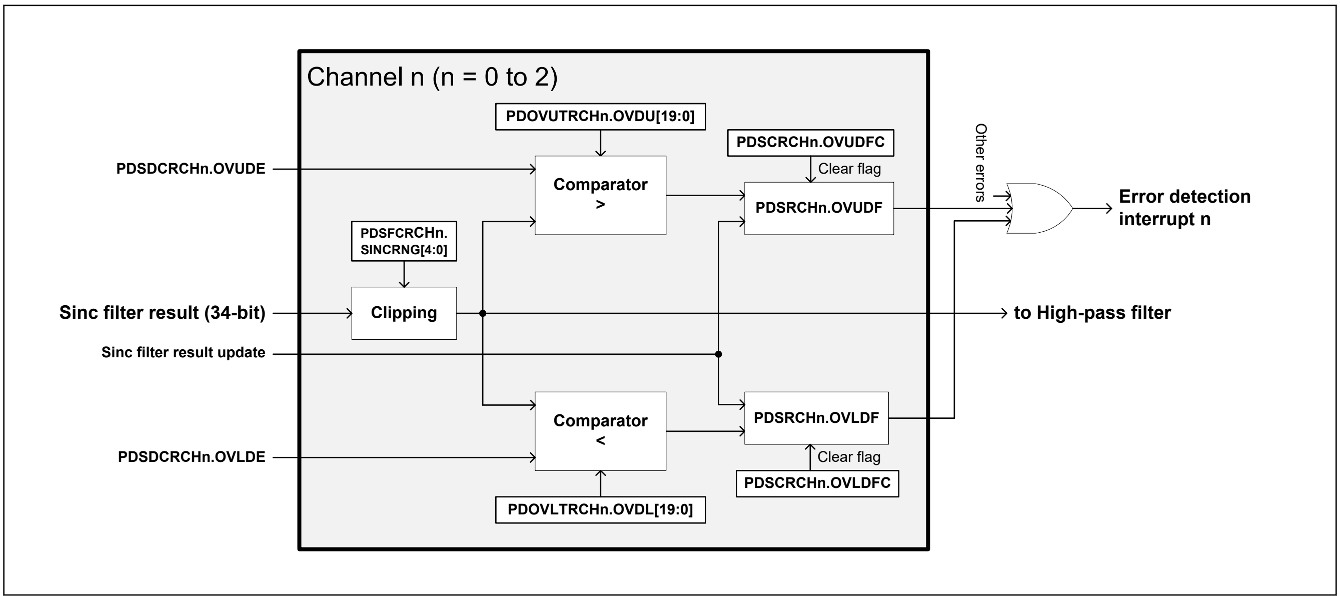


Figure 50.13 Overvoltage detection function

50.3.10 Error

Table 50.19 shows a PDM-IF error list.

Table 50.19 PDM-IF error list

No	Register name	Description	Comment	Function (Related section)
1	PDSRCHn.SCDF	Short circuit detection flag		section 50.3.8. Short-circuit Detection
2	PDSRCHn.OVLDF	Overvoltage lower limit detection flag		section 50.3.9. Overvoltage Detection
3	PDSRCHn.OVUDF	Overvoltage upper limit detection flag		section 50.3.9. Overvoltage Detection
4	PDSRCHn.BFOWDF	Buffer overwriting detection flag	For debugging	section 50.3.7. Data Buffer

No.4 errors are for debugging. Software must set registers and operate PDM-IF so that these errors do not occur. If any of these errors occurs, stop PDM-IF following section 50.4.2. Stop Flow and change the settings.

50.3.11 Interrupt Function

PDM-IF has the following interrupt function.

Table 50.20 PDM-IF interrupt list (1 of 2)

No	Signal name	Interrupt name	Number of signals	Pulse or Level	Related register bit name: Common register	Related register bit name: Channel register	Function (Related section)
1	PDM_DATn (n = 0 to 2)	Data reception interrupt CHn	3	Level	PDCSR.DRFn	PDSRCHn.DRF	section 50.3.7. Data Buffer
2	PDM_SDET	Sound detection interrupt (3-channel status flags are ORed.)	1	Level	PDCSR.SDFn	PDSRCHn.SDF	section 50.3.6. Sound Detection

Table 50.20 PDM-IF interrupt list (2 of 2)

No	Signal name	Interrupt name	Number of signals	Pulse or Level	Related register bit name		Function (Related section)
					Common register	Channel register	
3	PDM_ERRn (n = 0 to 2)	Error detection interrupt CHn	3	Level	PDCSR.EDFn	PDSRCHn.SCDF	section 50.3.8. Short-circuit Detection
						PDSRCHn.OVLDF	section 50.3.9. Overvoltage Detection
						PDSRCHn.OVUDF	section 50.3.9. Overvoltage Detection
						PDSRCHn.BFOWDF	section 50.3.7. Data Buffer

50.4 Setting Flow

Figure 50.14 and Figure 50.15 show examples of the main procedure of PDM-IF.

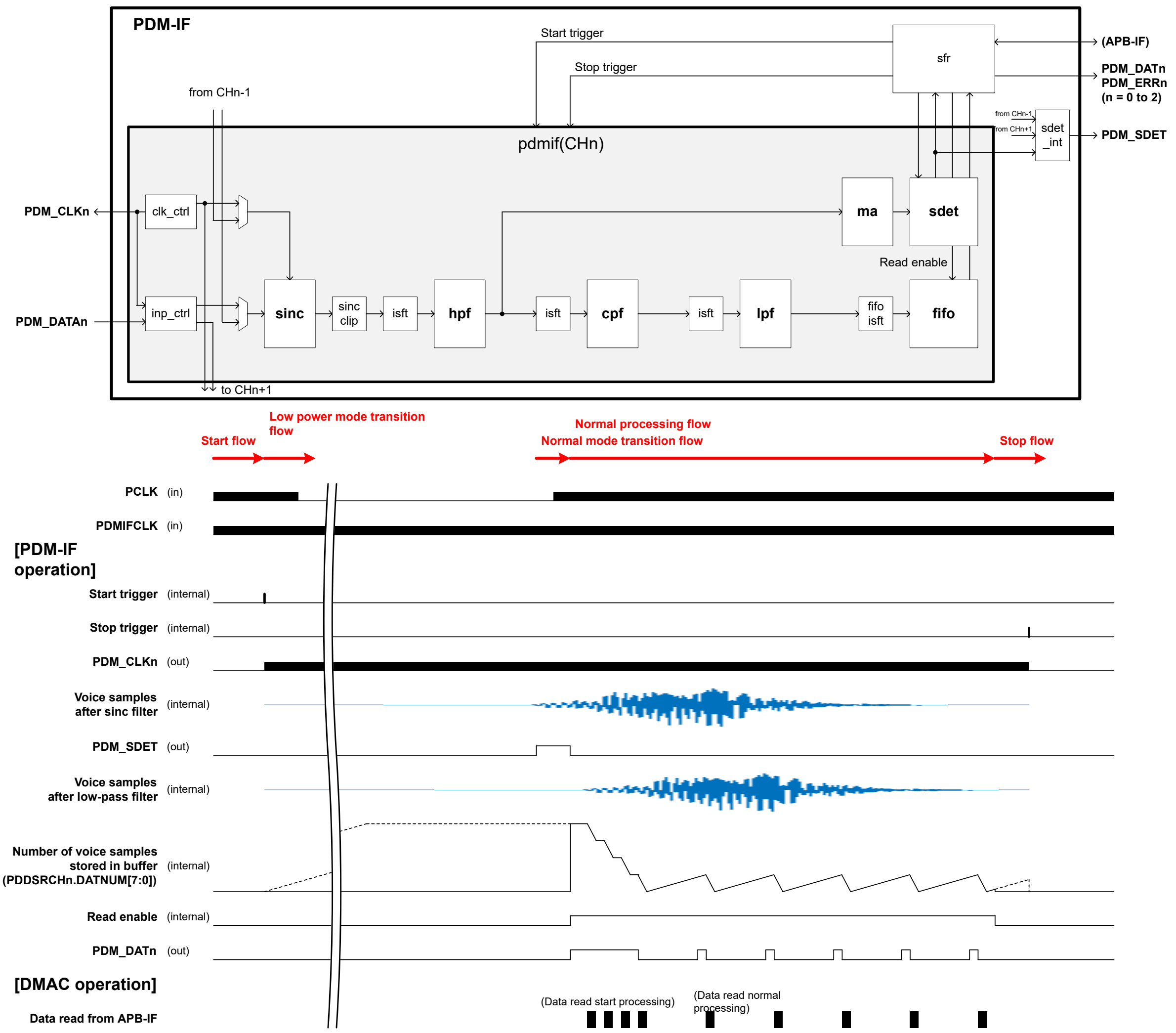


Figure 50.14 PDM-IF operation example

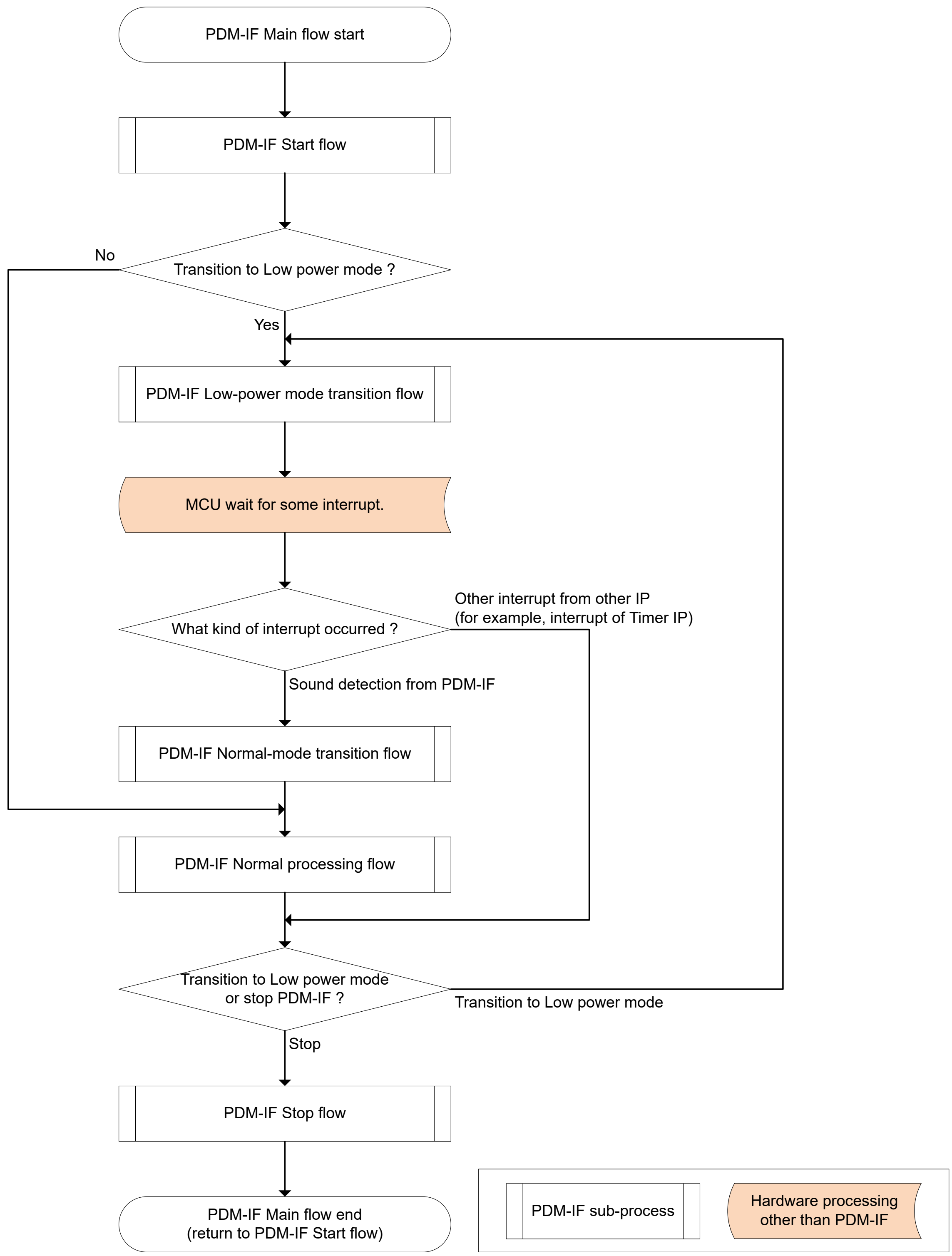


Figure 50.15 PDM-IF setting main flow with low-power mode transition

Figure 50.16 shows example of the main procedure without low-power mode transition.

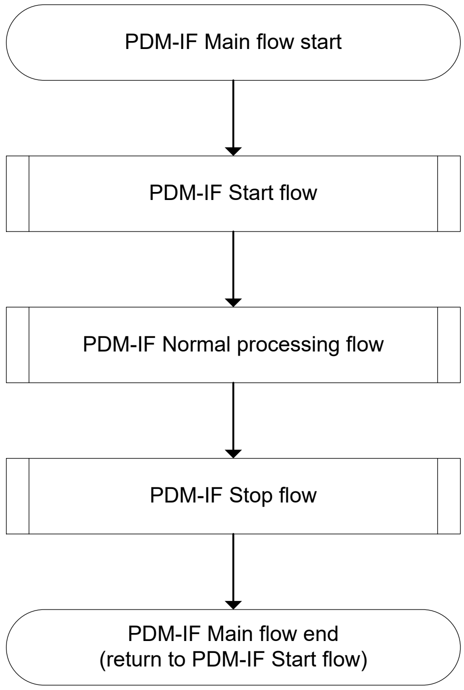
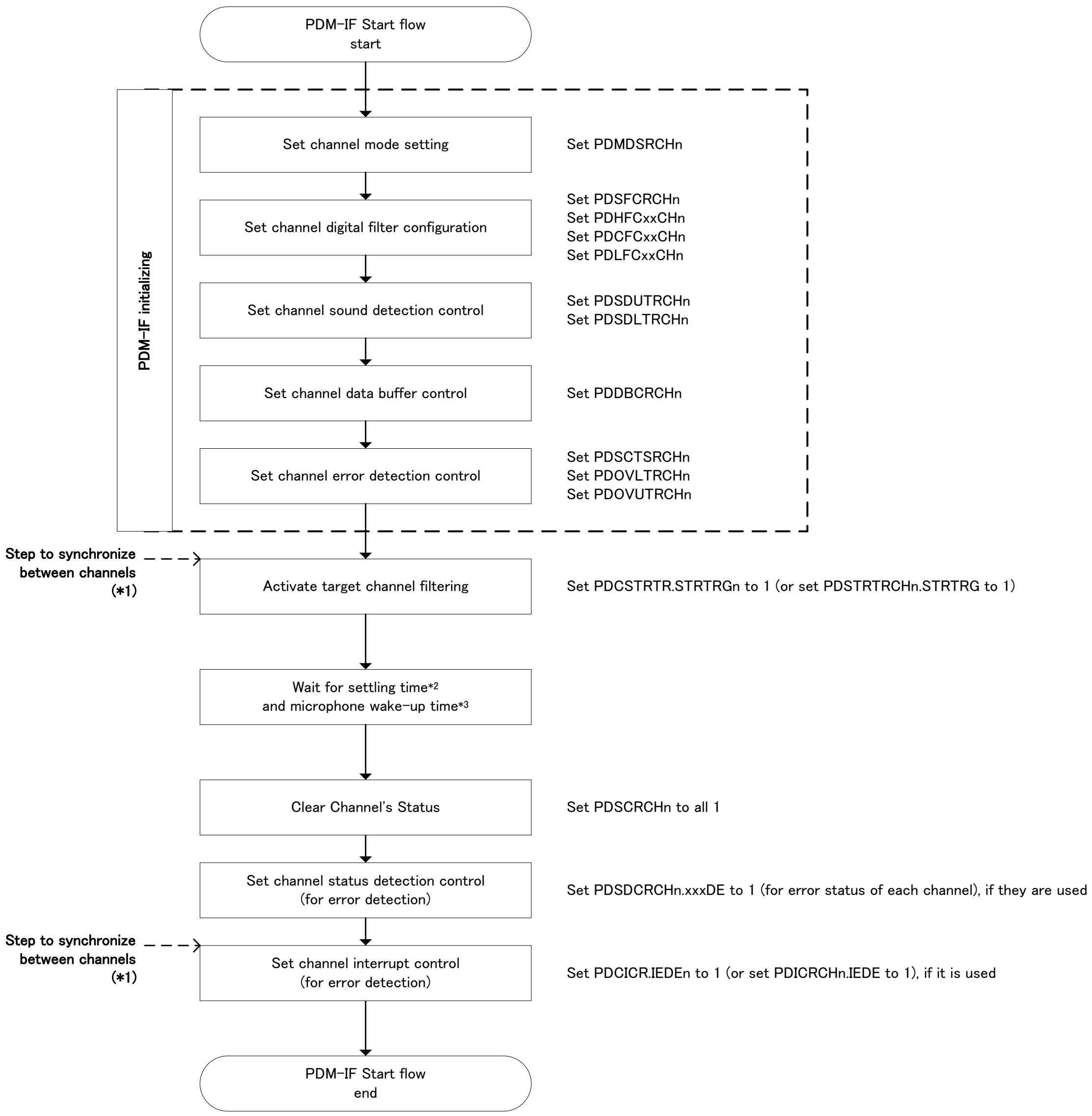


Figure 50.16 PDM-IF setting main flow without low-power mode transition

50.4.1 Start Flow

Figure 50.17 shows example of the start procedure of PDM-IF.

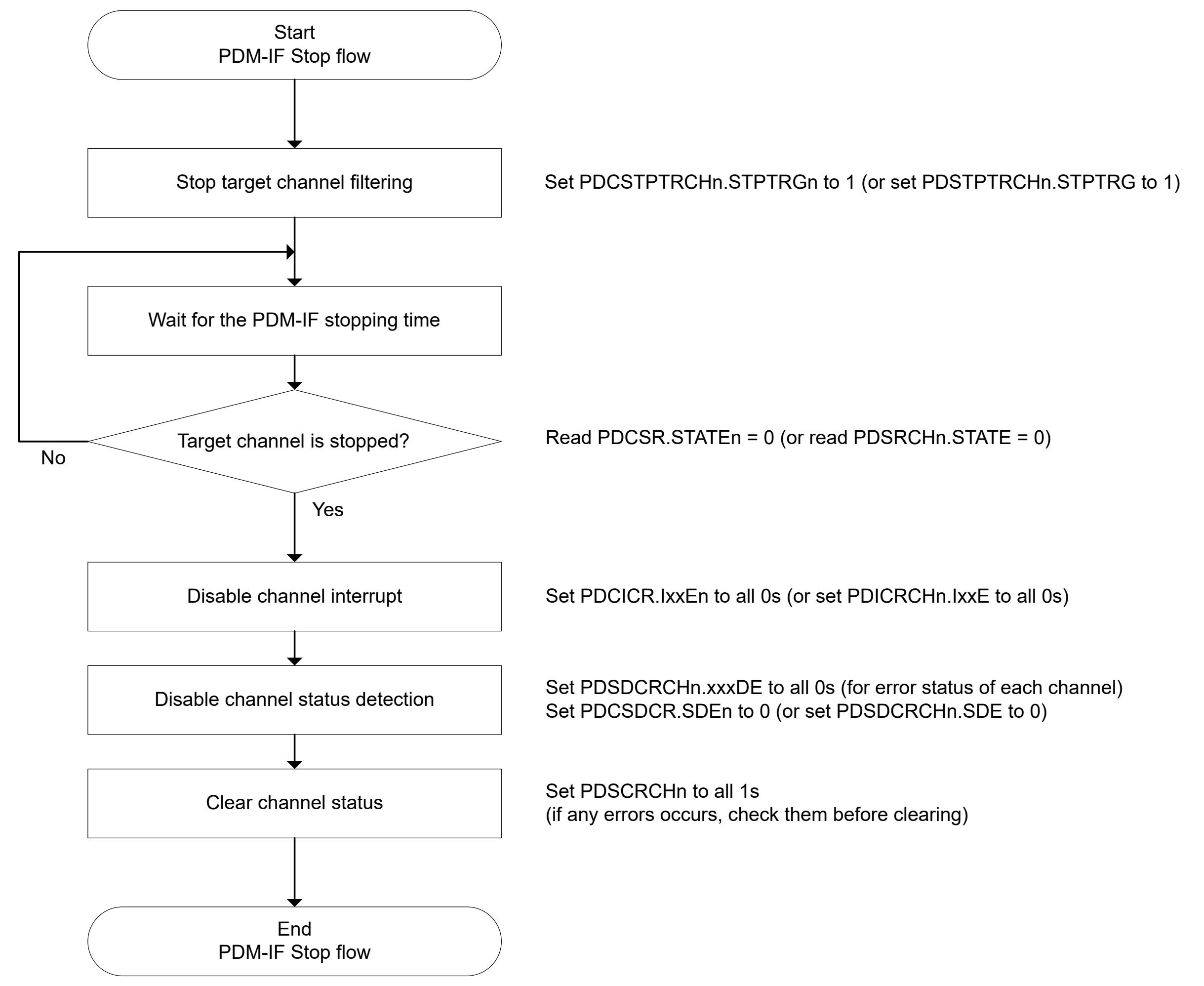


Note: xx or xxx refers to the register that matches the name pattern as similar part (ex: PDLFC) + different part (xx or xxx).
Note 1. If the channels are to be synchronized, complete the settings for each channels before this step.
Note 2. For details, see section 50.3.4.7. Settling Time of Channel Activation/Setting Change.
Note 3. Confirm the wake-up time from the connected microphone specifications.

Figure 50.17 PDM-IF start flow

50.4.2 Stop Flow

Figure 50.18 shows example of the stop procedure of PDM-IF.

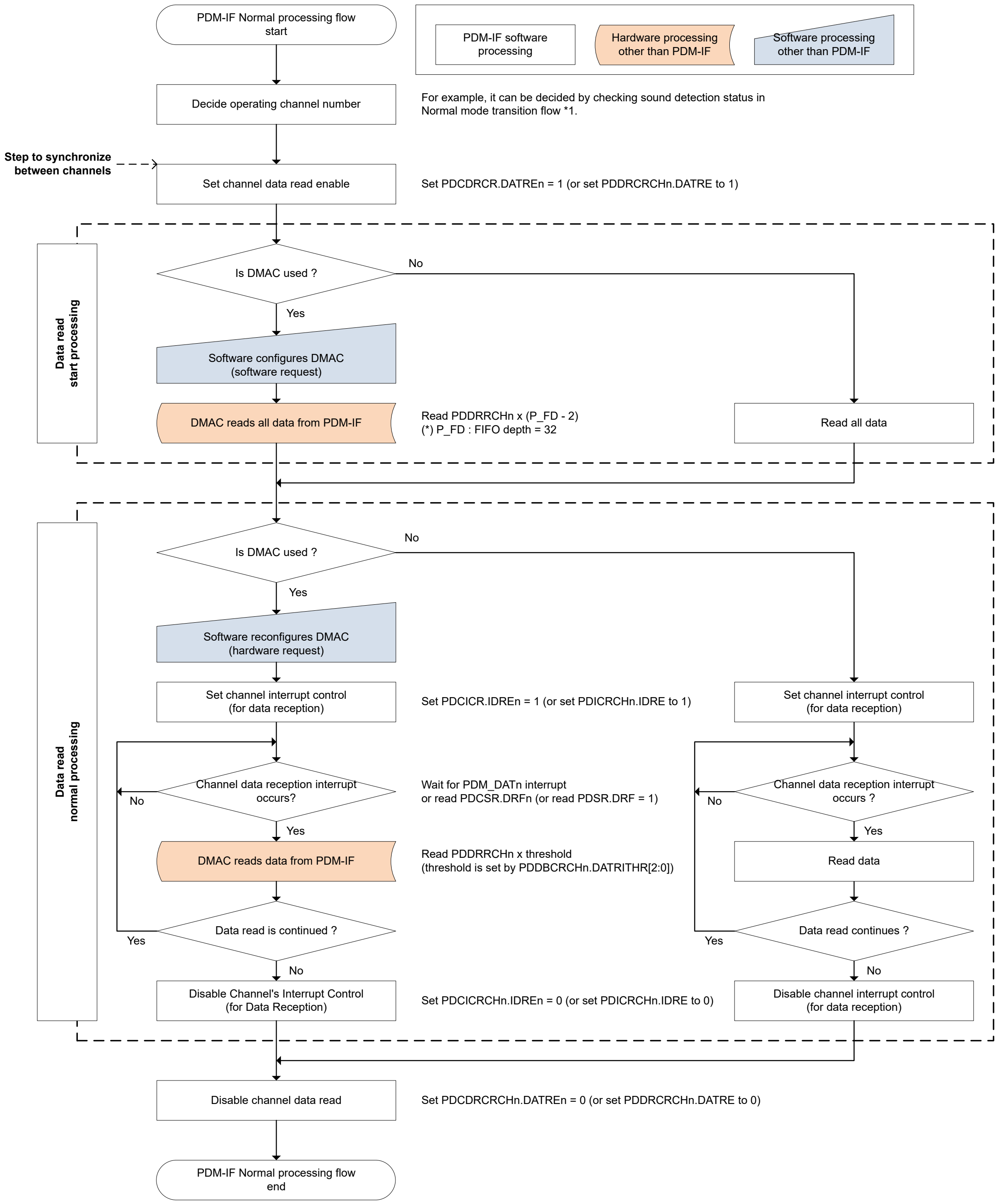


Note: xx or xxx refers to the register that matches the name pattern as different part (xx or xxx) + similar part (ex: DE).

Figure 50.18 PDM-IF stop flow

50.4.3 Normal Processing Flow

Figure 50.19 shows example of the normal processing procedure of PDM-IF.

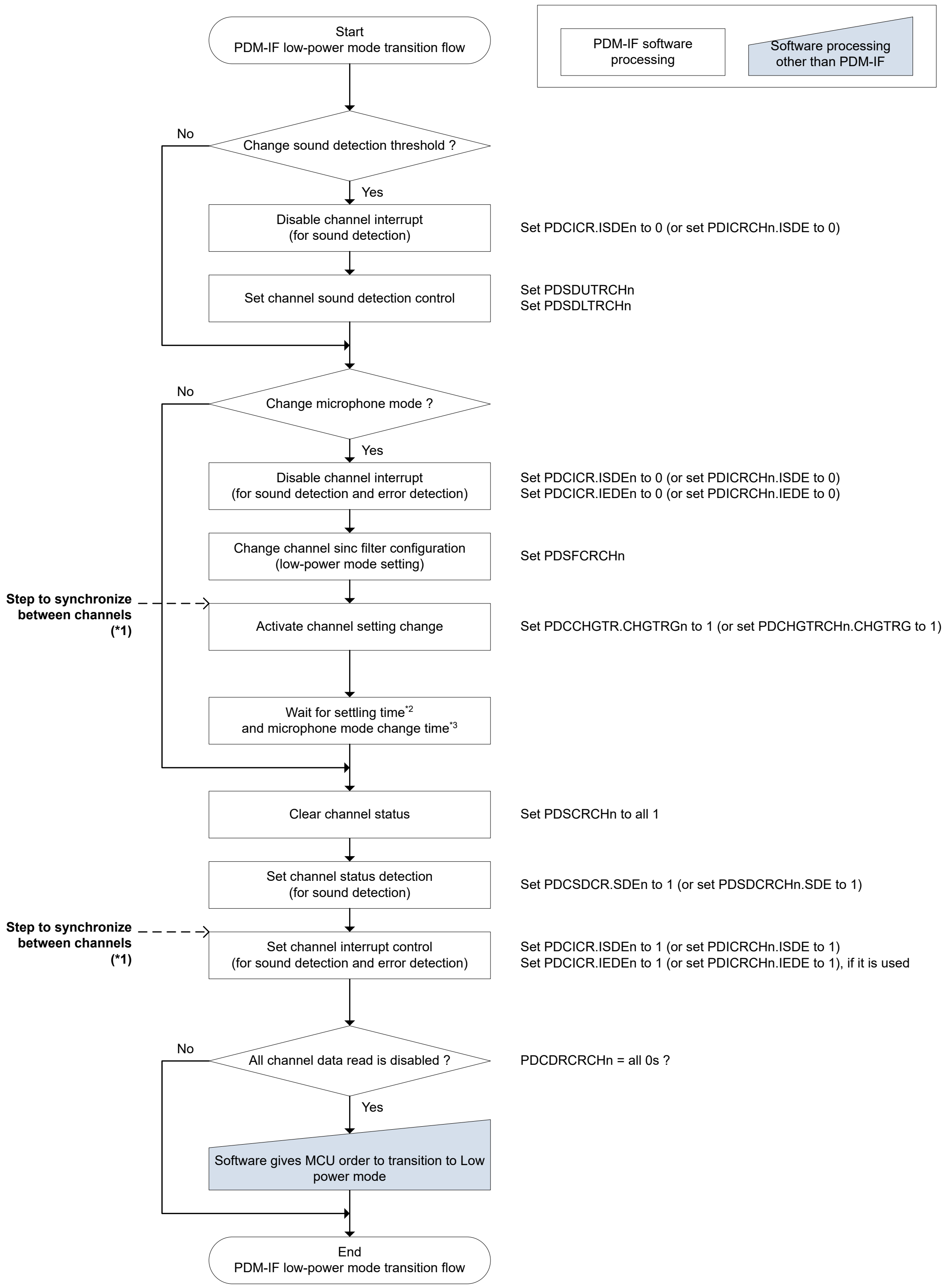


Note 1. For details, see Figure 50.21.

Figure 50.19 PDM-IF normal processing flow

50.4.4 Low-power Mode Transition Flow

Figure 50.20 shows example of the low-power mode transition procedure of PDM-IF.

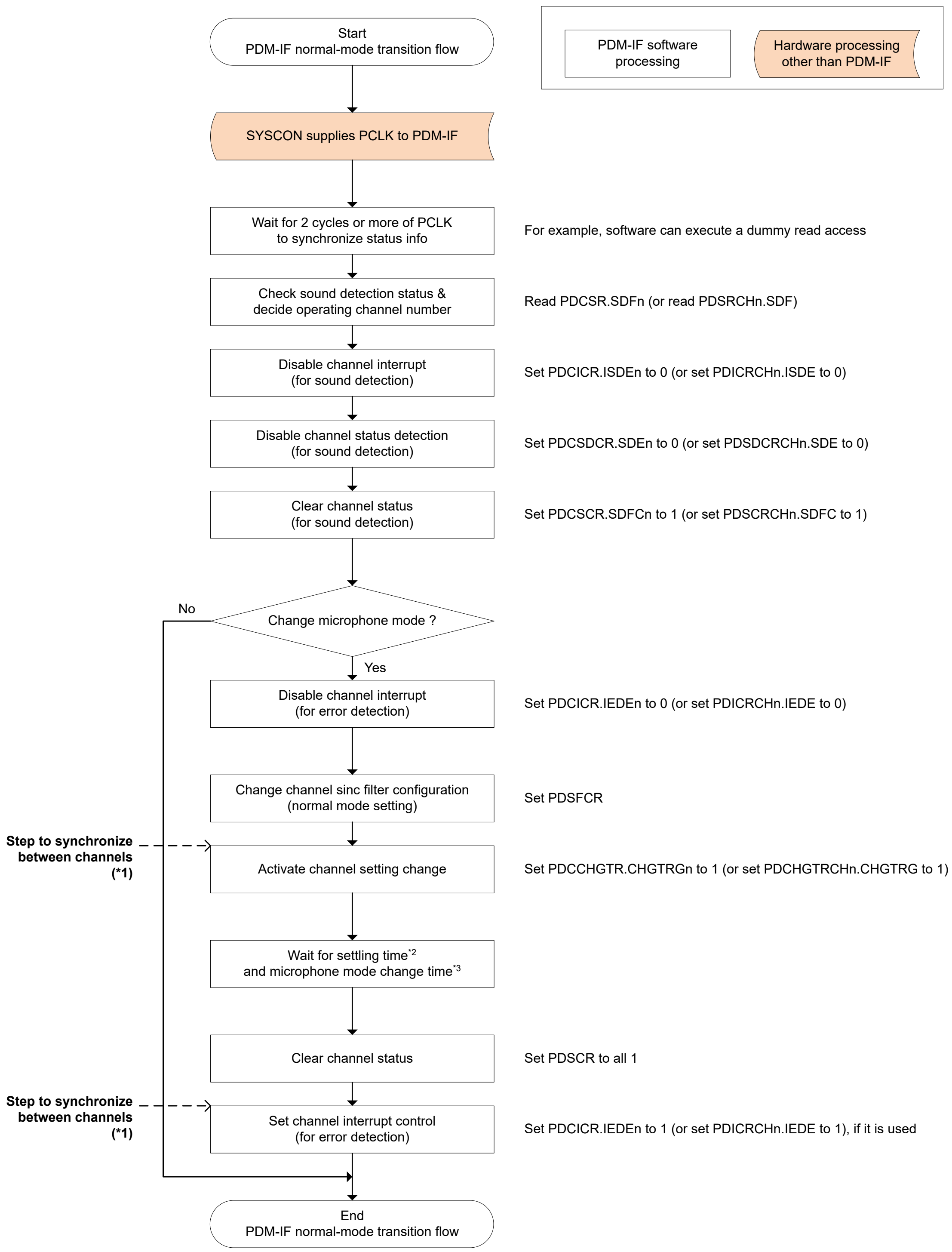


Note 1. If the channels are to be synchronized, complete the settings for each channel before this step.
Note 2. For details, see section 50.3.4.7. Settling Time of Channel Activation/Setting Change.
Note 3. Confirm the PDM microphone mode frequency and mode change time from the connected microphone specifications.

Figure 50.20 PDM-IF low-power mode transition flow

50.4.5 Normal Mode Transition Flow

Figure 50.21 shows example of normal-mode transition procedure of PDM-IF.



Note 1. If the channels are to be synchronized, complete the settings for each channel before this step.
Note 2. For details, see section 50.3.4.7. Settling Time of Channel Activation/Setting Change.
Note 3. Confirm the PDM microphone mode frequency and mode change time from the connected microphone specifications.

Figure 50.21 PDM-IF normal-mode transition flow

50.5 Usage Notes

50.5.1 Data Reception Interrupt Threshold Register Setting

A data reception interrupt PDM_DATn is issued as a level output. Therefore, if the number of samples in buffer still exceeds the data reception interrupt threshold set in PDDBCRCHn.DATRITHR[2:0] when data reading of DMA finishes, the interrupt controller might not detect the edge of PDM_DATn. The situation happens if data reading time exceeds sampling rate*1 × threshold. Set the appropriate threshold so that the interrupt controller can detect the interrupt.

In case the threshold cannot avoid the situation, read PDDBCRCHn.DATRITHR[2:0] and restart DMA by software after reading the data.

Note 1. Sampling rate: Time to spend to store a sample in buffer.

50.5.2 Frequency Characteristics of Filters with Default Settings

Figure 50.22 shows the frequency characteristics of the filters with the default settings for reference information.

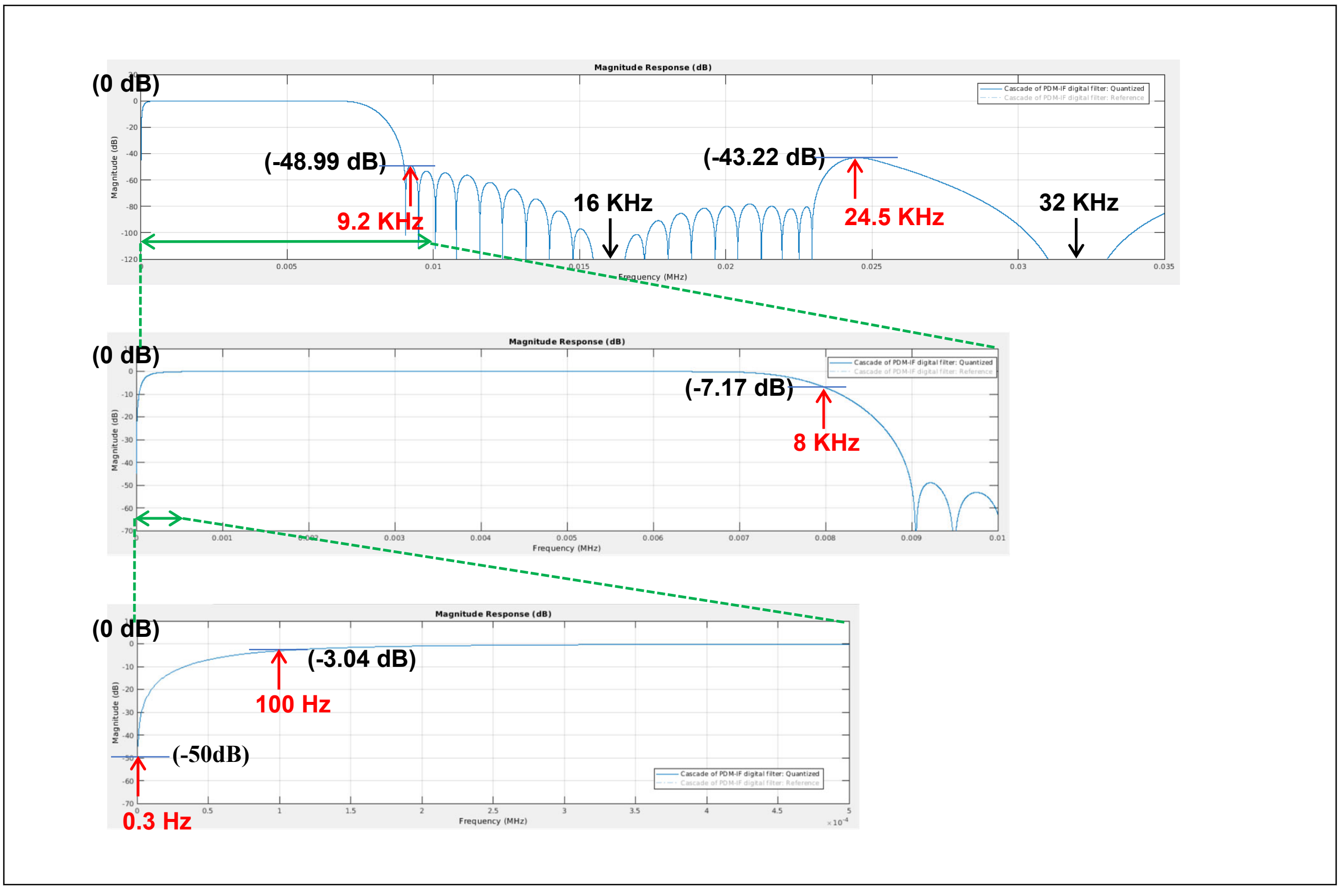


Figure 50.22 Frequency characteristics of filters with default settings

Table 50.21 Frequency characteristics of filters with default settings (1 of 2)

Frequency (Hz)		Magnitude (dB)
Cut-off frequency of LPF	24.5 kHz	-43.22
	…	…
	9.2 kHz	-48.99
	8 kHz	-7.17

Table 50.21 Frequency characteristics of filters with default settings (2 of 2)

Frequency (Hz)		Magnitude (dB)
Passband	7 kHz	-0.39
	6 kHz	-0.07
	…	…
	150 Hz	-1.63
	100 Hz	-3.04
Cut-off frequency of HPF	50 Hz	-7.02
	0.3 Hz	-50.00